

Optimizing Pharmaceutical Control with Multi-Task Contextual Bandits: Addressing Batch Heterogeneity for Improved Manufacturing Efficiency

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Problem definition: Efficient control is critical for continuous pharmaceutical manufacturing to ensure the production of high-quality tablets despite variations in raw material attributes and environmental factors. Furthermore, the data in *2024 MSOM Data-Driven Research Challenge* indicates that batches with the same observed attributes are heterogeneous in their impact on the tablet quality, i.e. there is hidden batch heterogeneity. We explore how to design optimal pharmaceutical control policies with limited data and hidden batch heterogeneity. **Methodology/Results:** We propose a contextual bandit model with related sequential batches to control a manufacturing process where the raw material batches have hidden heterogeneity. We propose a novel algorithm that combines results from the multi-task learning and the contextual bandit literature to minimize the regret in this setting. We show that the batch heterogeneity is best represented by a new multi-task random forest model. We extend the theoretical regret guarantees for both the Thompson Sampling and greedy algorithms to the setting of contextual bandits with related sequential batches. The results of four numerical experiments show that the multi-task random forest has good prediction performance, and the control policy recommended by our algorithmic framework outperforms the historical policy by as much as 20%. **Managerial implication:** In practice, overlooking hidden heterogeneity between batches can lead to significant loss in quality. Our framework is flexible to model the production process with limited but relevant data, offering a valuable tool for improving pharmaceutical manufacturing and related production problems.

Keywords : Multi-Task Learning, Contextual Bandit, Data Heterogeneity, Pharmaceutical Production

1. Introduction

In pharmaceutical production, continuous manufacturing provides significant advantages over traditional batch processing. It enables real-time monitoring and control, allowing manufacturers to adjust parameters dynamically in response to environmental and material variation, and improving product consistency and quality (Domokos et al. 2021). This approach also enhances efficiency and cost-effectiveness by reducing downtime and ensuring a smooth production flow with real-time signals and sensors. Precisely controlling manufacturing processes is essential to adhere to the regulatory standards set by governing

bodies, e.g., the US Food and Drug Administration (FDA 2024) and the International Council for Harmonisation Guidelines (ICH 2009).

Despite its benefits, efficient control in continuous manufacturing is challenging due to limited data and unknown process uncertainties between batches of refilled raw materials in the system. Batches that are identical in observed attributes can show hidden heterogeneity (Myerson et al. 2015). Consequently, one cannot pre-compute actions based solely on the observed attributes of the raw materials and the environmental variables. Instead, the control policy has to adapt to the batch using online data as well as historical data from related batches, i.e., one has to not only balance exploration (learning the characteristics of the batch) and exploitation (providing near-optimal actions) but also learn from related historical data.

We present a new data-driven algorithmic framework that controls a continuous production system efficiently with unobserved heterogeneity in raw material batches. The *MSOM Data-Driven Research Challenge* (Martagan et al. 2024), provides the testbed for our work. The MSOM challenge focuses on a continuous tablet manufacturing process on a commercial scale (Holman et al. 2021), consisting of four main stages: formulation, blending, compression, and coating with continuous manufacturing, and experiencing various sources of process uncertainty in the production pipeline. The two main challenges are as follows:

- (i) *Data Scarcity* : Optimal control policies for chemical processing require both an accurate system model and real-time sensor data for feedback control. Unfortunately, in practice, the data is very limited and only contains costs associated with given actions and states, e.g., only approximately 10 data points of tablet attributes per batch (Holman et al. 2021). A new data-driven approach is needed to discover the optimal action while maintaining reasonable production costs during the learning phase.
- (ii) *Data Heterogeneity* : The raw material properties vary significantly across batches, and therefore, the predictive power of model trained on previous batches is limited. Furthermore, the optimal decision for each batch, even under identical environmental conditions, depends on hidden (unobserved) attributes unique to each batch that must be learned. In Figure 1 we plot the within-batch (pooled) and out-of-batch performance of a random forest model trained on 40 batches. It is clear that the out-of-batch performance of the model is very poor. The results in Figure 1 clearly demonstrate

that accounting for heterogeneity is very important for out-of-sample accuracy. Ignoring heterogeneity leads to severe degradation in prediction performance, with the dropping from 0.15 to negative. The training details are in Section 4.2.

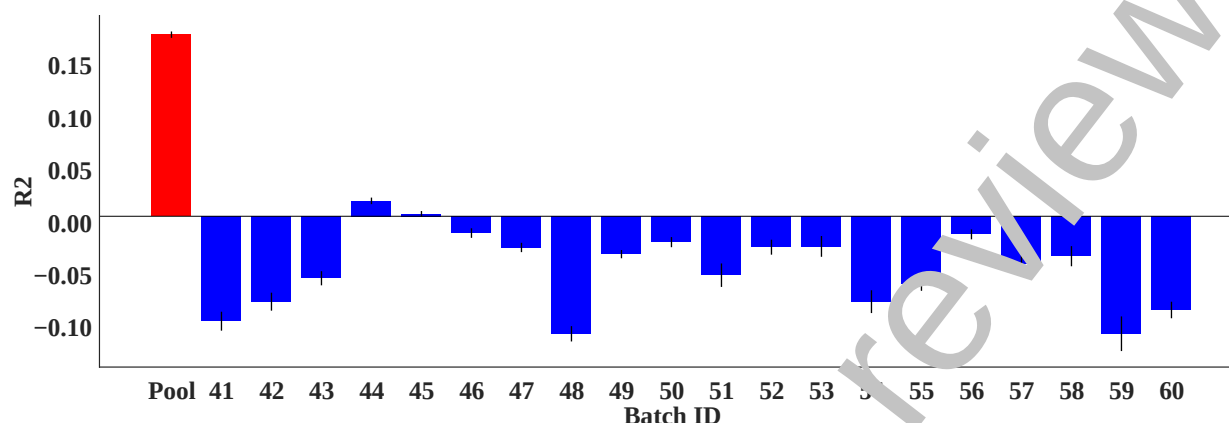


Figure 1 Out-of-batch performance of the prediction model for the tablet quality cost that does not take batch heterogeneity into account for the first 40 batches (40), where “Pool” means the out-of-sample performance within these 40 batches and the others refer to the IDs of batches that can be negative since the model transfers poorly to batches that are not in the training data. Here we set the beginning of each batch to the epoch when the active pharmaceutical ingredient (API) is refilled.

This result in Figure 1 implies that the prediction model needs to be adapted to each new batch by interacting with its data. In this paper, we propose a new prediction model that starts with a baseline estimate for each batch and then progressively refines it as more data from the batch becomes available. Building on this, we introduce a new decision framework that we call contextual bandit with related sequential batches to choose actions. Our results demonstrate that accounting for batch heterogeneity enhances predictive accuracy (see Section 6.2), and that the actions selected by the proposed bandit model lead to lower incurred costs (see Section 6.3). We build on multi-task learning (MTL) (Baxter 2000), more specifically the multi-task bandit literature, which proposes general algorithms to transfer knowledge between related problem instances (tasks) to improve predictive or prescriptive analytics. However, results in the literature either study the setup where each task is learned simultaneously (Yang et al. 2021, Duan and Wang 2023) instead of sequentially, as is the case in our setting, or under the linear model assumption (i.e., the setup in Bastani et al. (2022)). Our work bridges these gaps and proposes the contextual bandit with related sequential batches (MTL-Seq-CB model, and an algorithmic framework for

minimizing regret in this model. We show that our algorithmic framework can identify the shared structure and achieve near-optimal performance even when there is heterogeneity across tasks.

1.1. Main Contributions

Our main contributions are as follows:

- (a) **Model formulation:** We formulate a contextual bandit problem where related but heterogeneous batches arrive sequentially (Seq-CB). In contrast to the standard contextual bandit instance, here the decision-maker has the opportunity to learn from related batches encountered in the past. In the MSOM Data Challenge, we identify blending and compressing as two separate stages that fit into the framework, and heterogeneity refers to the hidden heterogeneity of raw materials for each batch.
- (b) **Algorithmic framework:** We propose the MTL-Seq-CB to solve the Seq-CB problem. When a new batch arrives, the decision-maker first solves the multi-task (representation) learning problem using the data from all the previous batches, and then calls a standard off-the-shelf contextual bandit algorithm. In the multi-task learning stage, we consider various embeddings that are flexible enough to handle the data heterogeneity. This includes an novel multi-task random forest for tabular data that is applicable to any multi-task learning setup.
- (c) **Performance guarantees:** We provide regret guarantees for the MTL-Seq-CB algorithm when the embedded contextual bandit problem is solved using Thompson Sampling or the greedy algorithm. Under mild conditions, we show that the asymptotic performance of both algorithms improves when multi-task learning is employed.
- (d) **Empirical results:** we validate our methodology in case studies from the pharmaceutical production pipeline. In particular, we find that multi-task random forest is able to effectively predict the hidden heterogeneity with out-of-sample R^2 being 0.35 and 0.56 in blending and compression stages respectively and compatible with their chemical dynamics. In the online decision-making, we implement our proposed methods in that pharmaceutical dataset and observe significant improvement over the historical policy. Solving the contextual bandit problem for each batch using either the Thompson Sampling or greedy algorithm results in a 20% improvement, on average, over the historical policy.

The rest of this paper is organized as follows. In Section 2 we review the related literature. In Section 3 we discuss the results of our data analysis that clearly show the existence of hidden batch heterogeneity. We describe the Seq-CB model in Section 4 and present bounds for our proposed MTL-Seq-CB framework in Section 5. In Section 6 we discuss the results of four numerical experiments, and the associated insights. In Section 7 we include some concluding remarks. Proofs and additional results can be found in the online appendix.

We use notations O , Ω and $\tilde{O}$ to describe the asymptotic performance of a function. Specifically, given two functions $f(T)$ and $g(T)$, $g(T)$ is said to be $O(f(T))$ and $f(T) = \Omega(g(T))$ if there exist positive constants c , and T_0 such that $g(T) \leq cf(T)$ for all $T \geq T_0$. Similarly, $g(T)$ is said to be $\tilde{O}(f(T))$ if $g(T)$ is $O(f(T) \cdot \log^k T)$ for some constant k .

2. Literature Review

2.1. Chemical Dynamics in Manufacturing Control

Conventional control methods for pharmaceutical manufacturing identify control policies for action selection on the basis of analytical models for the chemical and mechanical dynamics of the processing system as a function of the process parameters, environmental variables, e.g., temperature and humidity, and raw materials attributes. For example, in blending, the action is the screws speed (Kumar et al., 2014) that affects the uniformity of the downstream resulting flow; and in compression, the action is the compression force (Diab et al., 2021) that impacts the physical properties of the tablets (Moghtadernejad et al., 2018). Note that both these actions have nonlinear effects (Adeleye 2019). Among environmental factors, temperature and humidity have the largest impact on tablet quality (Prado and Vajta 2021). Moisture has been shown to affect the flowability of pharmaceutical excipients (Croucher and Briens 2014), adding unpredictability and uncertainty to the manufacturing process. The physicochemical properties of the ingredients explain variabilities in the production process, e.g., bulk density (Kreiser et al., 2023), order of mixing ingredients (Pinali et al., 2011). Despite these qualitative insights, the exact dynamics of each production pipeline are often unique and analytically challenging to identify. This underscores the need for data-driven operational techniques to analyze and optimize manufacturing processes on a case-by-case basis (Gamset al., 2014).

There is some existing work on integrating domain knowledge of chemical engineering applications with data-driven learning approaches for interpretable, reliable, and scalable decision-making (Venkatasubramanian 2019). Specifically, data-driven operations (Mišić

and Perakis 2020) play a crucial role in pharmaceutical production, e.g., optimizing real-time flow rates (Damiri et al. 2025), ensuring quality control (Singh 2024), and predicting tablet quality (Aksu et al. 2012). These approaches typically rely on offline supervised learning to identify effective control policies. We propose a new modeling approach for online data-driven decision making, that builds on the insights from contextual bandits and multi-task learning literature, and is tailored to address the challenges in continuous pharmaceutical production.

2.2. Contextual Bandits with Offline Data

Contextual bandit problem is a framework for online decision-making where the optimal action depends on some state at each decision epoch, with applications in healthcare (Tewari and Murphy 2017), dynamic pricing (Qiang and Bayati 2016) and electronic commerce (Li et al. 2010). We refer reader to Chapter 18 of Lattimore and Szepesvári (2020) for a detailed survey. Note that the contextual bandit approach has not been considered in pharmaceutical production literature.

In the standard contextual bandit problem (Lattimore and Szepesvári 2020), there is a single “task”, meaning that while the stochastic model governing the cost as a function of the state and action is unknown, it remains constant. Some studies investigate performance improvement when the contextual bandit algorithm is provided with offline historical information. These include (feature-based) dynamic pricing (Bu et al. 2022), adaptive A/B tests (Li et al. 2021), and best arm identification (Agrawal et al. 2023). This work assumes that the problem instances are identical in the offline and online stages. Recent work incorporates shifts between the offline and the online stage in simple settings such as multi-armed bandit with a change mean of each arm (Cheung and Lyu 2024) or dynamic pricing (Wang et al. 2024) (a change in the coefficients in the pricing model). However, the offline data is still iid, i.e., there are only two “tasks”: the historical task and the new task. We propose a model that accommodates heterogeneity in the offline dataset with multiple tasks, and a multi-task learning based framework to effectively learn from this heterogeneous historical data.

2.3. Meta and Multi-task Learning for Decision Making

Since each new batch is a new bandit (task) instance, we are in the *metalearning* setting where knowledge acquired from prior tasks can be used to efficiently address new

tasks (Finnetal.2017). In the bandit setting, the meta-Thompson Sampling algorithm uses empirical Bayes to set the Gaussian prior for new tasks (Nabi et al. 2022, Kveton et al. 2021). Variants of this approach have also been applied to dynamic pricing (Bastani et al. 2022). These problem setups are closely related to ours; however, in these setups the shared representation is known, and that significantly simplifies the problem.

The effectiveness of meta-learning depends on learning an embedding for transferring information between tasks. The goal of multi-task learning (Baxter 2000) is to improve generalization and performance across multiple related tasks when the task embedding is parametric. In the supervised learning setting (Maurer et al. 2016), by assuming the unknown task parameters are close in some metric (Bastani 2021), multi-task learning improves generalization by pooling the data for all tasks and then regularizing the task parameters (Tripuraneni et al. 2020, Du et al. 2021, Duan and Wang 2023). In the bandit setup, some works investigate the problem of minimizing the total regret in multiple parallel related bandit instances (Yang et al. 2021, Hu et al. 2021, Xu and Bastani 2025). In contrast, in our setting, the tasks or bandit instances arrive sequentially.

In other operational applications, there is work on improving the prediction performance for new products and services with limited data by leveraging historical data from other related products and services. This approach has been applied to ride-sharing (Qin et al. 2020), product sales forecasting (Hao et al. 2019, Baardman et al. 2018), and inventory control (Feng et al. 2023). We contribute to this emerging area by developing a model extending this meta-learning approach to continuous pharmaceutical production.

3. Exploratory Data Analysis

In this section, we describe the key process and challenges using the data analysis from *MSOM Data-Driven Research Challenge* 2024.

3.1. Summary Description of Pharmaceutical Process

We focus on new data provided by Holman et al. (2021) and synthesized by Martagan et al. (2024). The data is from a pharmaceutical production process where real-time monitoring of process parameters is provided by a GEACDC-50 control system. The system processes 6,000 kg of raw materials and manufactures over 15 million tablets over a period of 170 hours. The production process feeds all the raw materials into a blender using loss-in-weight (LiW) feeders that control the total amount and mixture of flows by measuring

their weight in real time. Five feeders deliver different raw material to the first blender to create a homogeneous mixture, with a lubricant added to the mixture into the second blender. Note that even though the production is continuous, the raw materials are fed in batches. Then, the system uses a compression step to process the blend into core tablets.

See Figure 2 for an abstraction of the production process that we use for our algorithm development. We only consider blending and compression stages as they are key stages in the process (Martagan et al. 2024). As discussed in Section 1, we model the process as a sequence of related contextual bandits. To this end, we next describe the state, the actions, and the associated costs.

We define the state of the system to be all the observable attributes of raw materials and the environmental variables. Note that the raw material attributes are constant for a given batch. The raw materials consist of Magnesium Stearate (MS), acetaminophen powder (as the Active Pharmaceutical Ingredient, API), lactose (L-16), Microcrystalline cellulose (Avicel), and Sodium Croscarmellose (Ac-di-sol). The latter three ingredients fed into the blenders do not show much variation except Avicel, and do not appear to impact the production outcome. Therefore, we only include MS, API, MS lot, and Avicel as raw material attributes in the state. The environmental variables include temperature and humidity.

The actions in our setting are screw speed in the blending stage, and the compression-related variables (compression force, and mill depth) in the compression stage. The quality of a tablet is defined by two attributes:

- (i) Blend potency: The concentration of the active components contained in a tablet and the portion of the tablets that are subjected to the out-of-specification alarm. We define the cost in the blending stage as mass flow times the total deviation of the blend potency and out-of-specification (OOS) concentrations summed over both the blenders. This cost captures all uncertainties in the blending stage, including the potency, prediction differences and OOS signals.
- (ii) Tablet physical attributes: The tablet's weight, hardness and thickness to achieve its golden profile (i.e., Table 13 of Holman et al. (2021)). We define the cost in the compression stage as the deviation of the tablet's hardness against its golden profile, since the other variables, i.e., weight and thickness, have much less variability.

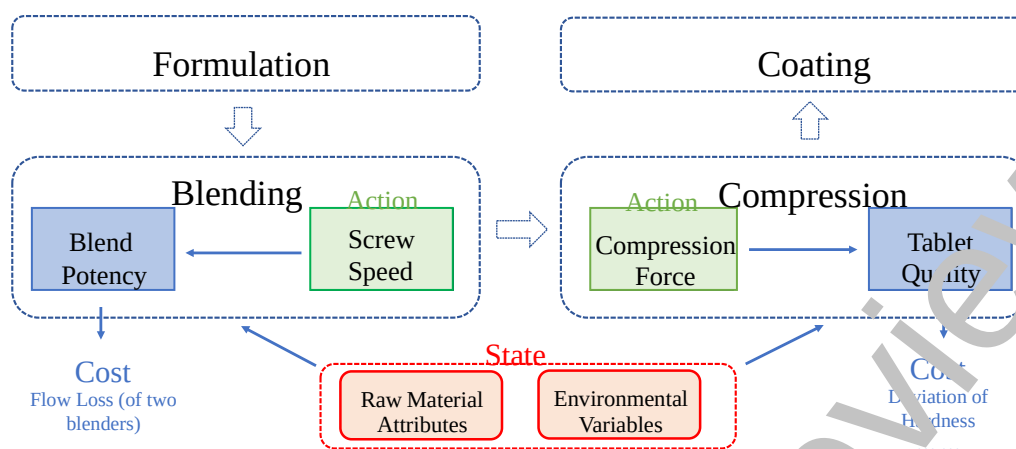


Figure 2 Summary of the pharmaceutical production process, highlighting the blending and compression stages.

We assume that the blending and compression stages can be modeled separately. Although these two stages might appear dependent, the operational range is broad enough to prevent significant coupling events. This separation assumption is supported by recent pharmaceutical literature (Pawar et al. 2016, Jakubowski and Ciepluch 2021). To empirically validate this separation assumption, we analyze the correlation between residuals of the “total flow loss” c_1 in the blending stage and the “deviation of hardness” c_2 in the compression stage after conditioning on shared state variables s . The results in Table 1 indicate that the residual costs, $c_1 - E[c_1|s]$ for blending and $c_2 - E[c_2|s]$ for compression, exhibit small correlations ($\rho < 0.1$). The correlations remain low (absolute values less than 0.15) even when we use other metrics for the costs c_1, c_2 .

Table 1 Correlation between outcomes in the blending and compression stages conditioned on state variables, given by $\text{Corr}(c_1 - E[c_1|s], c_2 - E[c_2|s])$, where $E[c_1|s]$, $E[c_2|s]$ are approximated using a random forest model. The boldfaced entries highlight the costs used in the subsequent analysis and their correlations.

Compression $c_2 \setminus$ Blending c_1	B-1 Loss	B-1 Flow Loss	B-2 Loss	B-2 Flow Loss	Total Flow Loss
Tablet Uniformity	0.115	-0.053	0.038	0.064	-0.094
Deviation of Hardness	-0.114	0.073	-0.046	-0.123	0.070

Notes. We denote the “B-1 loss” as the summation of the deviation of blend potency and OOS Concentration for blender 1, and “B-1 total loss” as its corresponding multiplication with the mass flow amount, “Sum Total Loss” denoting the B-1+B-2 total loss.

3.2. Interplay between States, Actions, and Costs

Next, we evaluate the impact of state variables, including raw material attributes, on the actions and costs in blending and compression stages. As a first step, we preprocess the

features and standardize the dataset. Next, we set the time indexing associated with the blending and compression stages to the epochs at which the cost for these stages is reported: 1 second for blending, and approximately 15 minutes for compression. These epochs may not be perfectly the epochs at which actions (and states) are updated. Therefore, we time-average the observed action (and states) to compute the actions (and states) associated with each epoch.

We fit a nonlinear random forest

$$\text{Action}_{i,t} = f(\text{Env}_{i,t}, \text{MS}_i, \text{API}_i, \text{Lot}_i) + \varepsilon_{i,t}^{(f)}. \quad (1)$$

to predict the historical actions $\text{Action}_{i,t}$ at each time point t of the i -th batch in the blending and compression stages, where $\text{Env}_{i,t}$ denotes the environmental variables including temperature and humidity, $\text{MS}_i, \text{API}_i, \text{Lot}_i$ denote the attributes of raw materials, i.e., the amount of MS and API, and their lots (category variables) at the i -th batch. Recall that we set the beginning of each batch to the epochs when API is refilled in Figure 1. Since MS is refilled at slightly different epochs, we compute the MS value associated with a batch by taking the time-weighted average of the MS values nearby. For example, suppose an API filling epoch is at 10 am. Therefore, a new batch begins at 10 am. Next, suppose the two nearest MS filling epochs are at 8 am with a fill of 8kg and at 11 am with a fill of 11kg, then the MS value associated with this batch is set to $\text{MS} = 8 \cdot \left(\frac{11-10}{11-8}\right) + 11 \cdot \left(\frac{10-8}{11-8}\right) = 10$.

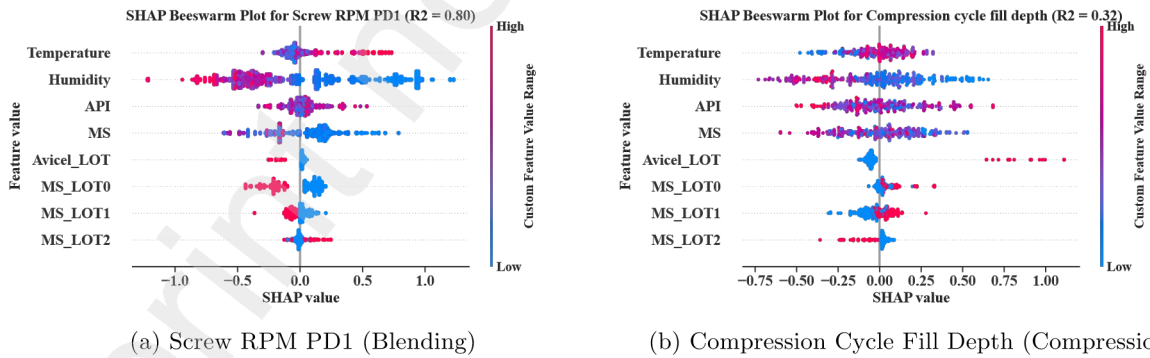


Figure 3 SHAP distribution plots for two specific actions, The results of other actions are deferred to Section EC.1. The beeswarm plot is an information-dense summary that highlights how the top features in a dataset impact the model output. Each instance is represented by a single dot on the row corresponding to each feature.

Figure 3 is the beeswarm plot of the Shapley values (Lundberg and Lee 2017) that shows the effect of each state variable in the random forest model to two selected actions on

the two stages. We observe that temperature and humidity exert a strong influence on decisions in each stage. Furthermore, the impact of variations in raw material properties, e.g., API, MS, and Avicellot, cannot be ignored across batches. For instance, changing the Avicellot impacts the actions in both the blending and compression stages, resulting in insignificant differences in compression fill depth, as illustrated in Figure 3b.

Next, we forecast costs for both the blending and compression stages as a function of actions and state via the random forest model:

$$\text{Cost}_{i,t} = g(\text{Env}_{i,t}, \text{MS}_i, \text{API}_i, \text{Lot}_i; \text{Action}_{i,t}) + \varepsilon_{i,t}^{(g)} \quad (2)$$

We report the feature importance of state and action variables in Figure 4. The results show that although actions are the dominant factors impacting cost, several state variables remain significant.

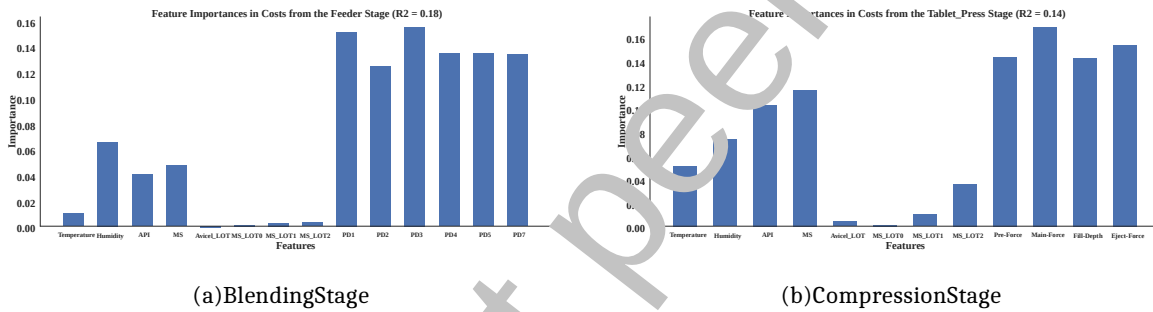


Figure 4 Feature importance of state and action variables for the cost

The results in Figures 3 and 4 demonstrate that the observed state variables impact the chosen actions, and both actions and state variables impact the cost in each stage.

3.3. Batch Heterogeneity

In this section, we establish that there is hidden batch heterogeneity, i.e., raw material batches with identical attributes have heterogeneous impacts on the action choice and cost.

In order to investigate whether the batch index has predictive information for historical actions and costs, we exhibit heterogeneity across batches, for (1) and (2), we incorporate the variability across batches via the following batch-specific random forest models:

$$\text{Action}_{i,t} = \sum_{j \in [J_f]} \theta_{ij}^{(f)} \phi_j^{(f)}(\text{Env}_{i,t}, \text{API}_i, \text{MS}_i, \text{Lot}_i) \quad (3)$$

$$\text{Cost}_{i,t} = \sum_{j \in [J_g]} \theta_{ij}^{(g)} \phi_j^{(g)}(\text{Env}_{i,t}, \text{API}_i, \text{MS}_i, \text{Lot}_i; \text{Action}_{i,t}), \quad (4)$$

Table 2 Results of incorporating batch heterogeneity in forest and linear models, where each entry in “Out-of-sample R^2 ” denotes model performance in (1) (forest-action), (2) (forest-cost) and (5) (linear-action), and each entry in its parentheses denotes the R^2 improvement including the batch-dependent term into the model (i.e., (3), (4) and (6) respectively). “#SS (1%)” and “#SS (10%)” denote the percentage of p-values smaller than 1% and 10% across $K = 63$ hypothesis tests with each null hypothesis being: $\mathcal{H}_{0,k} : \gamma_{1,k} = 0, \forall k \in [K]$.

	Forest Model	Linear Model		
Outcome Name	Out-of-sample R^2	Out-of-sample R^2	# SS(1%)	# SS(10%)
Response Variable: Action				
Screw RPM PD1	0.7477 (0.0799)	0.5351 (0.1786)	48.3%	68.3%
Screw RPM PD2	0.9099 (0.0539)	0.6354 (0.1051)	37.5%	62.5%
Screw RPM PD3	0.6471 (0.1504)	0.3245 (0.4229)	48.3%	61.7%
Screw RPM PD4	0.6896 (0.0361)	0.4043 (0.1614)	32.2%	49.2%
Screw RPM PD5	0.7993 (0.0913)	0.4169 (0.1954)	26.4%	43.4%
Screw RPM PD7	0.6492 (0.0640)	0.4624 (0.0969)	21.2%	48.1%
Pre-compression force	0.2696 (0.2771)	0.1393 (0.5413)	36.7%	73.3%
Main compression force	0.6802 (0.1457)	0.8191 (0.1588)	68.3%	73.2%
Compression cycle fill depth	0.2706 (0.2813)	0.2733 (0.3964)	21.1%	47.4%
Ejection force tablet	0.9082 (-0.0086)	0.6981 (0.1620)	8.3%	16.7%
Response Variable: Cost				
Blending Cost	0.1495 (0.1517)	-	-	-
Compression Cost	0.3346 (0.0638)	-	-	-

where $\phi_j^{(f)}$ (resp. $\phi_j^{(g)}$) denotes the output of the j -th decision tree of the random forest for $f(\cdot)$ (resp. $g(\cdot)$) in (1) (resp. (2)). This formulation reduces to the standard random forest when $\theta_{ij}^{(f)} = 1/J_f$ (resp. $\theta_{ij}^{(g)} = 1/J_g$) for each of the J_f (resp. J_g) trees in the action (resp. cost) prediction model for each batch i . We set $J_f = J_g = 20$ in the blending stage and $J_f = J_g = 5$ in the compression stage. See Section 4.2 for the training details. In the column marked “Out-of-Sample R^2 ” for the “Forest Model” in Table 2, we report the out-of-sample R^2 and R^2 -improvement (in parentheses) from incorporating batch heterogeneity. It is clear that incorporating batch heterogeneity improves prediction for all actions in the blending stage and most actions in the compression stage, and also the prediction of blending and compression costs significantly. In Appendix EC.1, we also visualize the 3D-projection of $\{\theta_i^{(g)}\}_{i \in [J_g]}$. That projection also exhibits large differences, implying their heterogeneity. These results clearly indicate that there is hidden heterogeneity in batches that is not fully accounted for by the observable attributes.

We also fit a linear model for each action in the blending and compression stages. Since the attributes of raw materials, e.g., API lot, MS lot, and the quantities of MS and API,

are functions of the raw material batch and have to be properly encoded, we cannot use the standard fixed-effects model (Angrist and Pischke 2009). We compare two standard regression setups, one that does not account for batch heterogeneity:

$$\text{Action}_{i,t} = \alpha + \beta^\top \text{Env}_{i,t} + \gamma_1 \text{MS}_i + \gamma_2 \text{API}_i + \gamma_3 \text{Lot}_i + \varepsilon_{i,t}, \quad (5)$$

and a second one that further accounts for batch-level heterogeneity:

$$\text{Action}_{i,t} = \alpha + \beta^\top \text{Env}_{i,t} + \gamma_1 \text{MS}_i + \gamma_2 \text{API}_i + \gamma_3 \text{Lot}_i + \left(\sum_{k \in [K]} \gamma_{1,k} D_{i,k} \right) \text{MS}_i + \varepsilon_{i,t}, \quad (6)$$

where $D_{i,k}$ is a dummy variable that equals 1 when $k = i$. In the standard model (5), the MS coefficient γ_1 is fixed across all batches, whereas in model (6), the coefficient $\gamma_1 + \gamma_{1,k}$ is different for each batch k . In the column marked “Out-of-Sample R^2 ” for the “Linear Model” in Table 2, we report the out-of-sample R^2 and R^2 -improvement (in parentheses) when using (5) and (6) respectively. For most actions in the blending and compression stages, we observe significant R^2 improvement, which agrees with the previous results in the random forest model. In the column marked “#SS (1%)” (resp. “#SS (10%)”), we report the percentage of p-values smaller than 1% (resp. 10%) for K separate hypothesis tests: $\mathcal{H}_{0,k} : \gamma_{1,k} = 0$ vs $\mathcal{H}_{1,k} : \gamma_{1,k} \neq 0$, $k \in [K]$. The higher the proportion of p-values that are small, the more significant the heterogeneity in the batches with respect to the quantity of MS fed into the system. In all actions, except ejection force, we observe over 40% of $\{\gamma_{1,k}\}_{k \in [K]}$ that are statistically significant at the 10% level and 20% of $\{\gamma_{1,k}\}_{k \in [K]}$ that are statistically significant at the 1% level. All these results emphasize that historical actions vary across batches even when conditioned on the same state variables that include all the relevant raw material attributes and environmental variables. Additional analysis that treats other variables, e.g., API, as batch-dependent, reveals similar patterns of heterogeneity. Note that we vary the coefficient of only one raw material attribute at a time. This is because $(\text{MS}_i, \text{API}_i, \text{Lot}_i)$ are the same across one batch i ; therefore, adding batch dependent coefficients for all three variables simultaneously is likely to result in over-identification, making it impossible to determine the exact values for each of these batch-dependent coefficients. A linear model for the cost prediction with or without heterogeneity has very low $R^2 (< 0.05)$; therefore, we do not report the results here.

In conclusion, the current actions taken and the underlying costs incurred in the system are influenced not only by the observed state variables, but also by some hidden factors

differing across batches beyond what is captured by existing data. Therefore, incorporating heterogeneous effects for each batch is crucial for achieving more effective control in pharmaceutical production.

4. Model

In this section, we describe the contextual bandit model with related sequential batches (**Seq-CB**). This model is a contextual bandit problem for each batch, and the sequence of batches are heterogeneous but related.

Setup. Denote a decision epoch by the tuple (i, t) , where i is the batch index and t is the time index in that batch. At each epoch (i, t) , the decision-maker observes the system state $s_{i,t} = (\xi_i, \eta_{i,t}) \in \Xi \times \mathcal{E}$ that consists of two components, where $\xi_i \in \Xi$ is a function of the batch i , and is the same for all epochs in a batch, and the “environment” state $\eta_{i,t} \in \mathcal{E}$ is a function of the epoch (i, t) , and can vary across epochs that belong to a batch. The decision-maker can take an action $x \in \mathcal{X}$, where $\mathcal{X}$ denotes the set of actions, which can be infinite and will be specified later in the case study. The cost $c_i(x, s_{i,t})$ that takes action x in state $s_{i,t} = (\xi_i, \eta_{i,t})$ in batch i satisfies the following assumption.

ASSUMPTION 1 (Conditional Data Generating Process). *We assume that the cost $c_i(x, s)$ for action x in state s in a given batch i is given by*

$$c_i(x, s) = \phi(x, s)^\top B \theta_i + \varepsilon, \quad (7)$$

where ε is sub-Gaussian noise with a known parameter σ , independent of x and s , θ_i is unknown where $\theta_i \sim N(\mu, \Sigma)$ with unknown $\mu \in \mathbb{R}^p$ and $\Sigma \in \mathbb{R}^{p \times p}$ sampled iid for each batch $i \in [K]$, $\phi: \mathbb{R}^{d_x} \times \mathbb{R}^{d_s} \mapsto \mathbb{R}^d$ is a known (possibly non-parametric) embedding vector, and the representation matrix $B \in \mathbb{R}^{d \times p}$ is possibly unknown. When B is unknown, $p \ll d$.

In Section 4.2, we show parametric and non-parametric embeddings used in practice that satisfy Assumption 1.

We assume access to historical data $\mathcal{D} := \{(s_{i,t}, x_{i,t}, c_{i,t})\}_{i,t=1}^{M,T}$ for M batches, each with T epochs from some historical policy.

ASSUMPTION 2 (Historical Control Policy). *For each batch $b \in [M]$ and $t \in [T]$, the environment state $\eta_{i,t} \sim \pi_1$, and the actions $x_{i,t} \sim \pi_{2,i}(\cdot | s_{i,t})$, $b \in [M], t \in [T]$, where both π_1 and $\pi_{2,i}$ are unknown. Note that $\pi_{2,i}$ can be batch dependent.*

Objective. Our objective is to design a policy whose expected cumulative regret

$$R_{K,T} = \sum_{k=1}^K \sum_{t=1}^T \phi(x_{M+k,t}, s_{M+k,t})^\top B \theta_{M+k} - \sum_{k=1}^K \sum_{t=1}^T \min_{x \in \mathcal{X}} \phi(x, s_{M+k,t})^\top B \theta_{M+k}, \quad (9)$$

is small over the K batches. Note that we index the batches by $M+k$, $k \in [K]$ in order to distinguish them from the M historical batches.

The contextual bandit model with related sequential batches is a good model for many issues in the pharmaceutical manufacturing control problem in the *MCOM Data-Driven Challenge* (see Section 3 for details). There, the state s represents the properties of raw materials, assumed to remain constant in a batch, and environmental factors, e.g., humidity and temperature, can vary over a batch. For the blending stage, the action x corresponds to the screw speed, and the cost c is defined as the deviation of the blend potency times the blending mass flow. For the compression stage, the action x denotes the compression force and fill depth, and the cost c is defined as the deviation of the tablet properties (e.g., hardness, weight) and the golden profile.

4.1. Algorithm Framework

Unlike the standard contextual bandit problem, which focuses on balancing the exploration-exploitation tradeoff over a single batch, we also need an effective strategy to learn across batches in our setting.

Algorithm. Our proposed MTL-Seq-C algorithm (see Algorithm 1) has two phases:

- (i) Multi-task learning phase (Line 2–Line 6): Solve a multi-task learning problem to compute improved estimates for the representation matrix B and the parameters θ_i for all the past batches, including the historical batch that is part of the dataset D . Note that we do not update the representation matrix B except in this phase. We also allow for the option where the algorithm does not learn B , and sets $B = I$.
- (ii) Contextual bandit phase (Line 7–Line 11): Solve a contextual bandit problem for the current batch using either the greedy algorithm (see Algorithm 2) or the Thompson Sampling algorithm (see Algorithm 3).

Greedy algorithm omits the exploration phase and may result in a suboptimal regret when the state does not exhibit sufficient randomness. However, this is not a concern for our use cases since temperature and humidity change frequently. When the state exhibits sufficient randomness, the greedy algorithm can achieve the optimal regret (Qian and

Bayati 2016, Bastani et al. 2021). Thompson Sampling algorithm explicitly encourages exploration through posterior sampling (Agrawal and Goyal 2013, Russo and Van Roy 2014), allowing it to update based on the evolving posterior distribution for optimization. To approximate the true prior with historical data in the Thompson Sampling algorithm, we initialize the Gaussian prior’s mean and covariance matrix with the empirical mean and covariance of the representation parameters estimated from the historical data.

Algorithm 1 MTL-Seq-CB Algorithm

Input: Historical data $\mathcal{D}$, embedding $\phi(\cdot, \cdot)$, $\text{RepMat} \in \{\mathbf{I}, \text{Unknown}\}$, $\text{BanditOracle} \in \{\text{Greedy}, \text{TS}\}$, regularization parameter $\lambda = c_0 p \sqrt{\frac{d}{T}}$.

1: **for** $k = 1, \dots, K$ **do**

2: **if** $\text{RepMat} = \text{Unknown}$ **then**

3:

$$(\hat{B}_{k-1}, \hat{\theta}_i, \hat{z}_i) \leftarrow \arg \min_{B, \theta_i, z_i} \underbrace{\sum_{i \in [M+k-1]} \frac{1}{T} \sum_{t \in [T]} (c_{i,t} - \phi(x_{i,t}, s_{i,t})^\top z_i)^2 + \lambda \|z_i - B\theta_i\|_2^2}_{=: f_i(B; \theta, z)} \quad (9)$$

4: **else**

5:

$$\hat{\theta}_i \leftarrow \arg \min_{\theta_i} \sum_{i \in [M+k-1]} \frac{1}{T} \sum_{t \in [T]} (c_{i,t} - \phi(x_{i,t}, s_{i,t})^\top \theta_i)^2. \quad (10)$$

6: **end if**

7: **if** $\text{BanditOracle} = \text{Greedy}$ **then**

8: Call Algorithm 2 ($\hat{B} = \hat{B}_{k-1}, \theta_0 = \frac{1}{M+k-1} \sum_{i=1}^{M+k-1} \hat{\theta}_i$)

9: **else if** $\text{BanditOracle} = \text{TS}$ **then**

10: Call Algorithm 3 ($\hat{B} = \hat{B}_{k-1}, \hat{\mu}_0 = \frac{1}{M+k-1} \sum_{i=1}^{M+k-1} \hat{\theta}_i, \hat{\Sigma}_0 = \frac{1}{M+k-1} \sum_{i=1}^{M+k-1} (\hat{\theta}_i - \hat{\mu}_0)(\hat{\theta}_i - \hat{\mu}_0)^\top$)

11: **end if**

12: Obtain the output from Algorithm 2 or 3 and combine them into $\mathcal{D}$

13: **end for**

4.2. Choices for Embeddings

We consider two different embeddings, i.e., choices for the function $\phi(x, s)$, that is an input to Algorithm 4.

- (1) Tree-based embedding: Tree-based models are known to perform well for tabular datasets (Grinsztajn et al. 2022), as is the case in our pharmaceutical setting. To account for batch heterogeneity, in Algorithm 4 we propose a novel tree-based model for multi-task learning. In Line 1, the algorithm generates a collection of d random

Algorithm 2 Greedy Algorithm

Input: Representation matrix $\hat{B}$, embedding $\phi(\cdot, \cdot)$, initial parameter θ_0

-
- 1: **for** $t = 1, \dots, T$ **do**

 - 2: Observe the state $s_t \sim \pi_1$

 - 3: $x_t = \arg \min_{x \in \mathcal{X}} \phi(x, s_t)^\top \hat{B} \theta_{t-1}$

 - 4: Deploy action x_t , and observe $c_t = \phi(x_t, s_t)^\top B \theta + \varepsilon_t$

 - 5: Update the parameter $\theta_t = [\sum_{i \in [t]} (\hat{B}^\top \hat{\phi}(x_i, s_i)) (\hat{B}^\top \hat{\phi}(x_i, s_i))^\top + \mathbf{I}]^{-1} \sum_{i \in [t]} c_i \hat{B}^\top \hat{\phi}(x_i, s_i)$.

 - 6: **end for**

Output: The data used for this batch $\{(s_t, x_t, c_t)\}_{t \in [T]}$.

Algorithm 3 Thompson Sampling Algorithm

Input: Representation matrix $\hat{B}$, embedding $\phi(\cdot, \cdot)$, mean $\hat{\mu}_0$ and covariance $\hat{\Sigma}_0$

- 1: **for** $t = 1, \dots, T$ **do**
- 2: Observe the state $s_t \sim \pi_1$
- 3: Sample parameter $\tilde{\theta}_t \sim N(\hat{\mu}_{t-1}, \hat{\Sigma}_{t-1})$
- 4: $x_t = \arg \min_{x \in \mathcal{X}} \phi(x, s_t)^\top \hat{B} \tilde{\theta}_t$
- 5: Deploy action x_t , and observe $c_t = \phi(x_t, s_t)^\top B \theta + \varepsilon_t$
- 6: Update the parameters of the posterior as:

$$\hat{\Sigma}_t = (\hat{\Sigma}_{t-1}^{-1} + (\hat{B}^\top \hat{\phi}(x_t, s_t)) (\hat{B}^\top \hat{\phi}(x_t, s_t))^\top)^{-1}, \quad \hat{\mu}_t = \hat{\Sigma}_t^{-1} \sum_{i \in [t]} c_i \hat{B}^\top \hat{\phi}(x_i, s_i)$$

- 7: **end for**

Output: The data used for the batch $\{(s_t, x_t, c_t)\}_{t \in [T]}$.

trees with $\phi(x, s) \in \mathbb{R}^d$ denoting the output of the trees. In Line 2, we minimize the loss while allowing the weights in each batch to vary from equal weights. Furthermore, we control the variation by the regularization parameter τ . It is clear that Algorithm 4 can be used to predict costs independent of the use of the embedding in Algorithm 1. Algorithm 4 outputs the embedding ϕ and cost prediction parameters $\hat{\theta}_i$, $i \in [M]$, for each batch in the input dataset. Specifically, the predicted cost for an action-state pair (x, s) in the i -th batch is given by $\hat{c} = \phi(x, s)^\top \hat{\theta}_i, \forall i \in [M]$.

This is similar to multi-task neural network (MTL-NN) where the task-independent representation $\phi(x, s)$ corresponds to all but the final shared layer, and the task-specific parameter θ plays the role of the last layer (Crawshaw 2020). A key distinction in the MTL-Forest is that $\phi(x, s)$ is defined by tree partitions rather than learned parametrized functions. Our proposed MTL-Forest is novel in its formulation com-

$$(2) \qquad \qquad \qquad N$$

$$\phi(x,s) = p_N(x,s) : \mathbb{R}^{d_x+d_s} \rightarrow \mathbb{R}^{N(x,s)}$$

$$\sum_{j \leq N} \frac{\prod_{k=0}^{j-1} (d_x+d_s+k)}{j!} \leq N \qquad \qquad \qquad d = N(x,s) =$$

$$x,s$$

pared to traditional multi-task parametric models. Additionally, as demonstrated by the results in Section 6.2, the MTL-Forest has the best empirical performance for the pharmaceutical production application.

Polynomial embedding: We also consider a degree- d polynomial embedding where $\phi(x, s)$, that consists of all polynomial combinations of the features with degree d . The polynomial dimension is given by $\binom{d + \dim(x, s)}{d}$. For example, if x and s are both one-dimensional, then the degree-2 polynomial features are $(1$

Algorithm 4 Multi-Task Random Forest (MTL-Forest)

Input: Data $\mathcal{D} = \{(s_{i,t}, x_{i,t}, c_{i,t})\}_{i,t=1}^{M,T}$, feature-ratio $\nu \in (0, 1)$, number of trees d , task similarity regularization parameter $\tau \geq 0$.

- 1: Fit d decision trees $\phi(x, s) = (\phi_1(x, s), \dots, \phi_d(x, s))^\top$ on $\mathcal{D}$ with c as output and (s, x) as input using $\nu \times 100\%$ random features for each tree.
- 2: Fit parameters by solving:

$$\arg \min$$

Output: The tree-based representation $\phi(x, s) \in \mathbb{R}^d$ and vector $\{\hat{\theta}_i\}_{i \in [M]}$

5. Theoretical Analysis

We now introduce some additional mild assumptions to help with the regret analysis.

ASSUMPTION 3 (Boundness of State and Action). *The embedding vector $\phi(x, s)$ satisfies $\|\phi(x, s)\|_2 \leq D < \infty$, for all x, s , and the representation matrix $\|B\|_F \leq D < \infty$.*

Recall from Assumption 1 that M denotes the number of historical batches, the parameter for each batch $\theta_i \in \mathbb{R}^p$, $b \in [M + K]$ and is sampled iid from $N(\mu, \Sigma)$.

ASSUMPTION 4 (Diversity of Representation). *$M \geq p + 1$ and $\lambda_{\min}(\Sigma) \geq \alpha > 0$.*

The parameter α quantifies the concentration of historical parameters where a larger α indicates greater diversity leading to more effective learning.

ASSUMPTION 5 (Diversity of State). *For any action $x \in \mathcal{X}$, the expectation $\mathbb{E}[B^\top \phi(x, \tilde{s})(B^\top \phi(x, \tilde{s}))^\top] \succeq c_1 \mathbf{I}$ over the random state $\tilde{s}$ for $c_1 > 0$.*

$$\hat{\theta}_i \in \bigcap_{\theta \in \Theta} \left\{ \sum_{t \in [T]} (c_{i,t} - \phi(x_{i,t}, s_{i,t})^\top \theta)^2 + \tau \|\theta - \mathbf{1}/d\|_2^2 \right\}, \quad i \in [M]. \quad (11)$$

Assumption 5 generally holds in our applications since the state is random here, and ensures that the associated empirical representation $\frac{1}{t} \sum_{i \in [t]} B^\top \phi(x_{j,i}, s_{j,i}) (B^\top \phi(x_{j,i}, s_{j,i}))^\top$ is full-rank with high probability when t is large for each batch j .

These assumptions imply the following regret bound for **MTL-Seq-CB**,

THEOREM 1 (Regret for the MTL-Seq-CB). *Suppose Assumptions 1, 2, 3 and 5 hold. Then the regret of MTL-Seq-CB over K batches satisfies the following bounds.*

- (a) *Representation matrix B is unknown, and Assumption 4 holds: The regret $\mathbb{E}[R_{K,T}] = \tilde{O}(pK\sqrt{T}) + \tilde{O}(p\sqrt{\frac{dKT}{M\alpha}})$ for the greedy algorithm, and $\mathbb{E}[R_{K,T}] = \tilde{O}(pK\sqrt{T}) + \tilde{O}(p\sqrt{\frac{dKT^3}{M\alpha}})$ for the Thompson Sampling algorithm.*
- (b) *Representation matrix $B = \mathbf{I}$: The regret $\mathbb{E}[R_{K,T}] = \tilde{O}(Kp\sqrt{T})$ for the greedy algorithm, and $\mathbb{E}[R_{K,T}] = \tilde{O}(Kp\sqrt{T}) + \tilde{O}(p\sqrt{KT^3})$ for the Thompson sampling algorithm.*

Note that when $B = \mathbf{I}$ is known, the regret bound does not depend on the offline data concentration α and the number of historic batches M . However in practice, the richness of the offline data helps choose the most effective representation from the model selection perspective. For reference, we compare the regret bound in Theorem 1 with the standard algorithm under the known representation and the correct prior for one task.

LEMMA 1 (Regret under Correct Prior and Task Representation). *Suppose Assumptions 1, 3 and 5 hold, and the representation matrix B and the exact prior of $\theta \sim N(\mu, \Sigma)$ are known. Then the regret $\mathbb{E}[R_{K,T}] = \tilde{O}(Kp\sqrt{T})$ for the greedy and Thompson Sampling algorithm.*

In the setting of Lemma 1, even with a correct prior and representation, each θ_{M+i} is unknown and needs to be learned independently. Consequently, the regret $R_{K,T}$ grows linear with K in Lemma 1. Comparing Theorem 1 and Lemma 1, we observe that the greedy algorithm achieves a better upper bound and closely approximates the true correct representation, with an additional regret term of $\tilde{O}(p\sqrt{\frac{dKT}{M\alpha}})$. This term is dominated by the inherent regret $\tilde{O}(Kp\sqrt{T})$ in Lemma 1. However, this does not imply that the greedy algorithm is theoretically better. In fact, these two algorithms are not directly comparable. The diversity condition (i.e., Assumption 4) favors the greedy algorithm to work well and ease the theoretical analysis (Bastani et al. 2021). Therefore, the regret guarantee of the greedy algorithm may not hold in other problem setups.

Improving regret bounds of the Thompson Sampling algorithm with respect to T in Theorem 1 remains a challenging problem. In related work, the meta Thompson Sampling algorithm (Kveton et al. 2021, Bastani et al. 2022), which leverages related tasks without requiring explicit representation estimation, also achieves a regret of order $\tilde{O}(\sqrt{KT} + \sqrt{KT^3})$. Ideally, the regret of such Thompson Sampling algorithms should also be similar to the greedy algorithm. However, achieving this bound in general settings (except the case when $K = \Omega(T^2)$) remains an open problem, and we leave closing this gap as our future work.

5.1. Proof Sketches

We need to address the following two key technical challenges to bridge offline estimation error bounds from standard results in offline multi-task learning, with the online regret bound established in Theorem 1:

- (i) Proposition 1 and Proposition 2 in Section 5.1.1 extend the error bound to the martingale setting while ensuring the misspecification error $\|\hat{B} - B\|$ and $\|\hat{\theta} - \theta\|$ is not large.
- (ii) Translating and analyzing these errors in the online setup without amplification compared with Lemma 1, leveraging novel Bayesian misspecification error techniques in Thompson Sampling algorithm, which is highlighted in Section 5.1.2.

We defer all proof technical details to Appendix EC.2.

5.1.1. Representation Error. In this section we state some key results to bound the estimation error of parameters. We focus on the case when B is unknown.

Proposition 1 (Estimation Error with Online Data). Suppose B is unknown and Assumptions 1, 2, 3, 4 and 5 hold. For each $k \in [K]$, let $(\hat{B}_{k-1}, \{\hat{\theta}_j\}_{j \in [M+k-1]})$ denote the solution of (9) in Algorithm 1. Then the following bound holds with probability at least $1 - \exp(-t)$, $\forall j \in [M+k-1]$:

$$\mathbb{E}_{M+k-1} \|\hat{B}_{k-1} \hat{\theta}_j - B \theta_j\|_2 \leq \sqrt{\frac{C_0 D^2 p^2}{\alpha}} \frac{d}{(M+k-1)T} + \frac{1 + \log(M+k-1) + t}{T},$$

where C_0 is some numerical constant.

When $k=1$, this result reduces to the bound in Theorem 4.5 in Duan and Wang (2023) since the data generating process of each of M batches is iid from Assumption 1. The key that generalizes the result to the online setting ($k > 1$) is to verify that Lemma B.6 in Duan

and Wang (2023) still holds even when the randomness of the loss gradient is taken into account. More specifically, this requires us to verify that the concentration bounds for a random matrix $G \in \mathbb{R}^{d \times (M+k-1)}$, where each column $G_i := \frac{1}{T} \sum_{t=1}^T (c_{i,t} - \phi(x_{i,t}, s_{i,t})^\top B \theta_i) B \theta_i$, $i \in [M+k-1]$ is the loss gradient, hold even when the data is dependent, i.e., a generalization of Theorem 5.39 in Vershynin (2010) to the martingale scenario.

The following result follow from Proposition 1 and the strong convexity of the loss in (9):

PROPOSITION 2 (Estimation Error of Representations and Parameters).

Suppose B is unknown. Then, for each $k \in [K]$, the solution of (9) in Algorithm 1 satisfies:

$$\max \left\{ \|\hat{B}_{k-1} - B\|_F^2, \max_{j \in [M+k-1]} \left\{ \|\hat{\theta}_j - \theta_j\|_2^2 \right\} \right\} \leq \kappa(H_{M+k-1})(\mathcal{E}_{M+k-1})^2,$$

where $\kappa(H_{M+k-1})$ is the conditional number of the matrix $H_{M+k-1} := \nabla_{B; \{\theta_j\}_{j \in [M+k-1]}}^2 F_{M+k-1}(B; \{\theta_j\}_{j \in [M+k-1]})$, with $F_{M+k-1}(B; \{\theta_j\}_{j \in [M+k-1]}) := 1/(M+k-1) \sum_{j \in [M+k-1]} \mathbb{E}_{\{\varepsilon_{j,t}\}_{t \in [T]}} [f_j(B; \theta_j, B\theta_j)]$ and f_j defined in (9).

When $B = \mathbf{I}$, the error of $\max_{j \in [M+k-1]} \left\{ \|\hat{\theta}_j - \theta_j\|_2^2 \right\} = \tilde{O}(p/T)$ from the standard results of linear regression (Hsu et al. 2012a).

5.1.2. Decision Error. In this section, we use the error bounds in Proposition 1 and Proposition 2, to establish regret bounds for the case when B is unknown.

Greedy Algorithm. In the greedy algorithm for the first task, we can decompose the regret into terms associated with the representation error $\|\hat{B}_0 - B\|_F$ and estimation error $\|\hat{\theta}_{M+1,t} - \theta_{M+1}\|_2$:

$$\mathbb{E}[R_{1,T}] \lesssim \sum_{t=1}^T \|\hat{\theta}_{M+1,t} - \theta_{M+1}\|_2 + T \|\hat{B}_0 - B\|_F \leq p \left(\sum_{t=1}^T \frac{1}{\sqrt{t}} + T \sqrt{\frac{d}{M\alpha T}} \right) = \left(1 + \sqrt{\frac{d}{M\alpha}} \right) p\sqrt{T},$$

where $\hat{\theta}_{1,t}$ represents the parameter estimate after t time periods for the first batch. Since Assumption 5 holds, the covariance matrix is well-conditioned, ensuring that the estimation error at each time satisfies $\|\hat{\theta}_{M+1,t} - \theta_{M+1}\|_2 \leq \frac{p}{\sqrt{t}}$. For $R_{K,T}$, following the same analysis, we obtain: $\mathbb{E}[R_{K,T}] = \sum_{k=1}^K \left(1 + \sqrt{\frac{d}{(M+k-1)\alpha}} \right) \sqrt{T} = pK\sqrt{T} + p\sqrt{\frac{dKT}{M\alpha}}$, where the second term arises due to the representation error.

Thompson Sampling Algorithm. To analyze the regret bound for Thompson Sampling, we borrow the following result for the error of Thompson Sampling under a misspecified prior:

LEMMA 2 (Informal Property of Thompson Sampling). *Suppose the true prior of the unknown parameter $B\theta$ is $\mathbb{P}_w$. Let $R_{1,T}(w)$ (and $R_{1,T}(w')$) denote the regret for one batch under the Thompson algorithm with priors $\mathbb{P}_w$ (and $\mathbb{P}_{w'}$) respectively. Then:*

$$R_{1,T}(w') - R_{1,T}(w) \leq CT^2 \cdot TV(\mathbb{P}_w, \mathbb{P}_{w'}).$$

for some constant C .

This result follows from Theorem E.1 in Simchowitz et al. (2021) by verifying that Algorithm 3 belongs to the class of 1-Monte Carlo algorithms. To apply this result in our setting and connect it with Lemma 1, we define the true prior $\mathbb{P}_w$ as the Gaussian distribution $N(B\mu, B\Sigma B^\top)$, where $\theta \sim N(\mu, \Sigma)$ follows the true prior and B represents the true representation structure. Similarly, we define the misspecified prior w' as the Gaussian distribution $N(\hat{B}\hat{\mu}, \hat{B}\hat{\Sigma}\hat{B}^\top)$ where $\hat{\theta} \sim N(\hat{\mu}, \hat{\Sigma})$ and $\hat{B}$ represents the estimated representation structure. This formulation ensures that the similarity between priors directly impacts regret in the online decision-making, provided that we can establish a bound on $\|\hat{B}\hat{\theta} - B\theta\|_2$. Furthermore, from Proposition 1 the above analysis holds for any $K > 1$.

We comment that when the representation matrix B is known, existing algorithms from Bastani et al. (2022) based on noise alignment can be applied. Specifically, they consider the event where $\hat{\mu}_t = \mu_t$ and $\hat{\Sigma}_t = \Sigma_t$ for $t \geq t_0$, where μ_t, Σ_t ($\hat{\mu}_t, \hat{\Sigma}_t$) denote the posterior mean and covariance matrix given the true (estimated) prior at the time t . Under this event, Algorithm 3 aligns with the Thompson sampling algorithm given the true prior for all $t \geq t_0$. However, when B is unknown, the noise alignment proof no longer holds due to the presence of $\hat{B} \neq B$. Even if $\hat{\mu}_t \hat{B} = \mu_t B$ at some time t , there is no guarantee that the relationship will hold at $t + 1$. This limitation necessitates an alternative analysis to bound the additional regret.

5.2. Implications of Theorem 1

We discuss the implications of our regret results in Theorem 1 here. First note that when B is known, a variant of Thompson Sampling algorithm from Bastani et al. (2022) yields $\tilde{O}(K\sqrt{T} + \sqrt{KT^3})$ regret bound in the dynamic pricing setup, which is the same as the guarantee in Theorem 1. Besides this simple case, our regret bounds are new.

The power of representation and diversity: In the case where B is unknown, if the new batch number $K \gg T$, the second term in the regret of each algorithm (i.e., $\tilde{O}(\sqrt{\frac{KT}{M\alpha}})$ in the greedy algorithm and $\tilde{O}(\sqrt{\frac{KT^3}{M\alpha}})$ in the Thompson Sampling algorithm) is dominated by $\tilde{O}(Kp\sqrt{T})$. As a result, the total expected regrets simplify to $\mathbb{E}[R_{K,T}] = \tilde{O}(Kd\sqrt{T})$, matching the regret bound in Lemma 1 with known representation and correct priors. Furthermore, in this case, if we forgo the use of any historical data D and directly learn the parameter $z_b = B\theta_i$ for that batch, since $z_b \in \mathbb{R}^d$, the total regret becomes $\tilde{O}(Kd\sqrt{T})$ (Russo and Van Roy 2014). Since $p \ll d$ when B is unknown (see Assumption 1), this regret bound is much larger than the $\tilde{O}(Kp\sqrt{T})$ regret achieved by our approach. This clearly illustrates the benefit of the representation to map the unknown parameter into a low-dimensional space. Furthermore, when the historical batches are more diverse (a large α) or the number of historical batches is large (a large M), the required number of new batches K that ensures $\mathbb{E}[R_{K,T}] = \tilde{O}(Kp\sqrt{T})$ decreases.

The power of offline data: Assumption 4 requires $M \geq p + 1$ to ensure that the representation error in Proposition 1 is controllable. When M is small, the representation error $\|\hat{B} - B\|_F$ can be large. In this scenario, due to the lack of a well-defined representation during the first $p - M + 1$ “test” batches, we incur an additional regret of the order $\Omega(\min\{(p - M + 1)T, (p - M + 1)d\sqrt{T}\})$, where the second term arises from a contextual bandit algorithm while ignoring B and directly learning z_b with dimension d . Here, offline data serves as a warm-start so that the regret of the first several batches is not large.

6. Case Study

We test the proposed methods on the data from a continuous manufacturing production line for pharmaceutical tablets (Martagan et al. 2024) to evaluate the benefits of different representation methods in cost prediction models (Algorithm 4) and online decision-making (Algorithm 1).

6.1. Setup

We aggregated data from different sources within the dataset (see Section EC.3.1 for details), which involves raw material, feeding, blending, compression and final tablet quality datasets. We refer to Martagan et al. (2024), Holman et al. (2021) for a detailed discussion of all the data and their summary statistics. Regardless of the stage type, the state variables include variables related to raw materials including total amount of the raw materials (API,

MS) that vary across batches, and environmental variables including the temperature and humidity. Each data entry represents the state, action, and cost for a single time period within a specific batch arrival, extracted from one of 62 different batches of raw materials arrivals. In the blending stage, the final filtered dataset consists of 21,389 entries, with each batch including from 31 to 1080 entries, and the cost is calculated as the total amount of mass flows subject to out-of-specification inspection and deviations in potency from its golden profile for the two blenders; In the compression stage, the final filtered dataset consists of 371 entries, with each batch including from 2 to 11 entries, and the cost is calculated as the absolute deviance of the hardness from its golden profile.

6.2. Evaluation of Offline Multi-Task Learning

We first consider the problem of developing a good prediction model for the blending and compression stages in the offline mode. We demonstrate that the MTL-Forest model proposed in Section 4.2 outperforms embedding methods based on polynomials. Specifically, we compare MTL-Forest against the polynomial representation in three different settings:

- (1) MTL: The training procedure in Section 4.2 used with either polynomials or trees as the base learners.
- (2) Pool: All the dataset is pooled, and a single polynomial or forest model is trained, ignoring batch distinctions.
- (3) Ind: A separate forest or polynomial model for each batch independently.

Details of the hyperparameters for both forest and polynomial models are provided in Section EC.3.2. We evaluate these models on two tasks, using an 80-10-10 data split, where we use 80% for training, 10% for validation (used to find the best hyperparameters), and 10% for testing within each batch. The averaged R^2 performance on the test dataset across 50 random splits is reported in Table 3.

Table 3 Out-of-sample R^2 of Testing Dataset for different methods in MTL, where the value to the right of indicates the standard error of cost, averaged over 50 different training/test splits

Setting	MTL-Forest	Pool-Forest	Ind-Forest	MTL-Poly	Pool-Poly	Ind-Poly
Blending	0.347 ± 0.003	0.216 ± 0.003	0.325 ± 0.003	0.193 ± 0.003	0.038 ± 0.002	0.130 ± 0.003
Compression	0.556 ± 0.029	0.395 ± 0.030	0.458 ± 0.052	-0.032 ± 0.563	-0.229 ± 0.553	0.277 ± 0.149

We find that MTL-Forest consistently outperforms both forest models trained with other training criteria, as well as all polynomial models. This indicates that MTL-Forest effectively captures batch heterogeneity and leverages tree-based representation for improved

predictive performance. MTL-Poly model performs relatively well for the blending stage; however, it performs very poorly for the compression stage, likely due to limited number of samples per batch and overfitting.

Besides the good empirical performance, we then study the optimal actions $\arg\min_{x \in \mathcal{X}} \phi(s_{i,t}, x_{i,t})^\top B \theta_i$ implied by MTL-Forest for batch i at time $t \in [T]$. Since the relationship $(i, s_{i,t}) \rightarrow x_{i,t}^* = \arg\min_{x \in \mathcal{X}} \phi(s_{i,t}, x_{i,t})^\top B \theta_i$, we fit another random forest to approximate $x_{i,t}^* = \zeta(s_{i,t}; i)$ with the dataset $\{(i, s_{i,t}, x_{i,t}^*)\}_{i,t=1}^{M,T}$, and interpret the sensitivity of actions based on the domain knowledge of pharmaceutical production. The results of this analysis, summarized in Table 4, indicate that the key variables are the amount of MS, API, humidity, and temperature, which are consistently important across all outcomes due to the following dynamics:

Table 4 Most impactful variables for blending and compression stages from the random forest interpretation

Decisions	Important Variable (besides batch heterogeneity)
Screw RPM PD1	API, Humidity, MS
Screw RPM PD2	MS, API, Humidity
Screw RPM PD3	API, MS, Humidity
Screw RPM PD4	MS, Humidity, API, Temperature
Screw RPM PD5	MS, Humidity, API, Temperature
Screw RPM PD7	MS, API, Humidity
Pre-compression force	MS, Humidity, API, Temperature
Main compression force	API, MS, Humidity, Temperature
Compression cycle fill depth	API, MS, Humidity, Temperature
Ejection force tablet	API, MS, Temperature, Humidity

- (i) The MS, API amount: In the blending stage, the MS and API amount highly influences screw speeds across all feeders. This is consistent with the fact that MS is a hydrophobic lubricant that improves the flowability of the powder via reducing powder flow resistance (Lemonen et al. 1992, Wan et al. 2010), and that API has significant moisture sensitivity and poor compressibility (DiMartino et al. 1996), which affects powder compression and flow behavior and compression respectively. In the compression stage, the water-repelling properties of MS affect interparticle bonding and compression strength subsequently.
- (ii) Humidity and temperature: In the blending stage, humidity and temperature have varying relative impacts across screw speeds in each feeder, with the most significant effects for PD1, PD4 and PD5. This is because Avicel (PD1, PD4) and Ac-di-sol (PD5)

are both hygroscopic, meaning higher humidity or temperature levels affect particle surface properties and flow properties. In contrast, other raw materials, L316 (PD2), API (PD3), MS (PD7) are stable and show relatively low-temperature sensitivity. In the compression stage, elevated temperatures increase the required compression force as API degrades at high temperatures, reducing cohesion and leading to tablets with suboptimal hardness. The remaining raw materials remain stable for the given temperature ranges. Humidity also plays an important role in compression force, because one of raw materials, Ac-di-sol, is highly hydrophilic and hygroscopic. By absorbing moisture, it increases the porosity of the powder mix, resulting in inconsistencies in the required compression force.

Overall, the interpretations in both blending and compression stages based on the chemical dynamics support the reasonability of our fitted model.

6.3. Evaluation of Online Decision Making

Next, we evaluate the policy implied by the prediction models. We only focus on MTL-Forest for the compression stage, since that was the only competitive prediction model (see Table 3). For the blending stage, we consider MTL-Poly and MTL-Forests since they were both competitive. We assumed the first $M = 4$ batches of raw materials as the historical data D , and evaluated performance over the subsequent $K = 20$ batches.

Action Update in a Batch. We update the action x_t at discrete time points since frequent changes may be restricted by regulatory and stability constraints, and the state variable such as humidity and temperature are typically updated at discrete time steps. For the blending stage, we update every 30 (resp. 5) sampling intervals (each interval is 1 second) for MTL-Forest (resp. MTL-poly) model. We set MTL-Poly with smaller sampling intervals since it involves larger uncertainty as B and θ are both unknown. However, our results are robust to different sampling intervals. For the compression stage, we update for every sampling interval, since the sampling interval for this stage is longer, approximately 15 minutes.

Representation Hyperparameters. We set the hyperparameters for the random forests as follows: number of trees = 20 (resp. 3), max depth = 10 (or 3), and the feature ratios as 0.5 for each tree in the blending (resp. compression) stage. We used degree-3 polynomial embedding, and set the dimension $p = 20$.

Other Benchmarks. We compare the performance of our approach with the following three policies:

- (i) Historical Policy: this policy uses the actions chosen in the dataset.
- (ii) MTL-only Policy: Use a fixed θ throughout each new batch (i.e., no update to θ in Algorithm 1).
- (iii) Oracle Policy: Assuming θ and B are known before each new batch starts, providing the upper bound of the policy performance.

When B is unknown, we also compare our approach with “*-Fixed” policy that does not update the estimator of $\hat{B}$ and $\hat{\theta}$ in Algorithm 1 when new batches arrive (i.e., skip the update of (9) after $K = 1$).

In Table 5, we report the average cost $\frac{1}{KT} \sum_{k \in [K]} \sum_{t \in [T]} \phi(x_{M+k,t}, s_{M+k,t})^\top B \theta_{M+k}$ of different policies on the “test” batches – recall that the historical policy uses actions recorded in the data set. Our findings indicate that, for both blending and compression stages, regardless of whether B is known or unknown, the performances of the greedy and TS policies are close to the oracle policy, and they outperform all other benchmarks, particularly the historical policy. When B is unknown, both Thompson Sampling and the greedy algorithms outperform their fixed counterpart (*-Fixed policy), indicating the power of representation updates.

Table 5 Average costs of different policies across new batches, where the boldfaced value in each column denotes the one with the smallest cost except the oracle. The value to the right of $\pm$ indicates the standard error of cost, averaged over 10 different training/test splits

Setting	B	Proposed Policies		Benchmark Policies				
		Greedy	TS	Oracle	MTL-only	History	Greedy-Fixed	TS-Fixed
Blending	Known	5.67 $\pm$ 0.12	5.56 $\pm$ 0.14	4.01 $\pm$ 0.15	6.21 $\pm$ 0.15	6.73	-	-
Compression	Known	0.53 $\pm$ 0.02	0.62 $\pm$ 0.02	0.40 $\pm$ 0.03	0.63 $\pm$ 0.03	0.87	-	-
Blending	Unknown	5.40 $\pm$ 0.22	5.29 $\pm$ 0.23	3.98 $\pm$ 0.24	6.15 $\pm$ 0.25	6.73	5.78 $\pm$ 0.25	5.86 $\pm$ 0.25

In Figure 5, we plot the histogram of the relative improvement (i.e., reduction) in cost between the historical policy and the MTL-Forest based Greedy and TS policies. To assess the statistical significance of the relative improvement, we conduct t -tests across all batches. Specifically, let $\bar{c}_{M+i}(\text{Policy})$ (resp. $\bar{c}_{M+i}(\text{History})$), $i \in [K]$ represent the average costs of a test policy and the historical policy over the K new batches. The t -value for the relative cost improvement is defined as $t_{\text{Policy}} = \frac{\text{mean}((\bar{c}_{M+i}(\text{History}) - \bar{c}_{M+i}(\text{Policy})) / \bar{c}_{M+i}(\text{History}))}{\text{s.e.}((\bar{c}_{M+i}(\text{History}) - \bar{c}_{M+i}(\text{Policy})) / \bar{c}_{M+i}(\text{History}))}$, where $\text{mean}(\cdot)$

and $\text{e}(\cdot)$ denote the mean and standard error of each sequence. The corresponding p -values are reported in the figure legends. Across both settings, we observe small p -values, indicating their statistical significance. Moreover, in most batches, our proposed policies achieve approximately 20% improvement over historical policies. While in some batches, our policies yield higher costs due to the limited number of available datapoints in that batch, these batches only consist of a small fraction of the total batches. We also conducted tests where we randomized over the training/test split and observed similar results (see Section EC.3.4 for details).

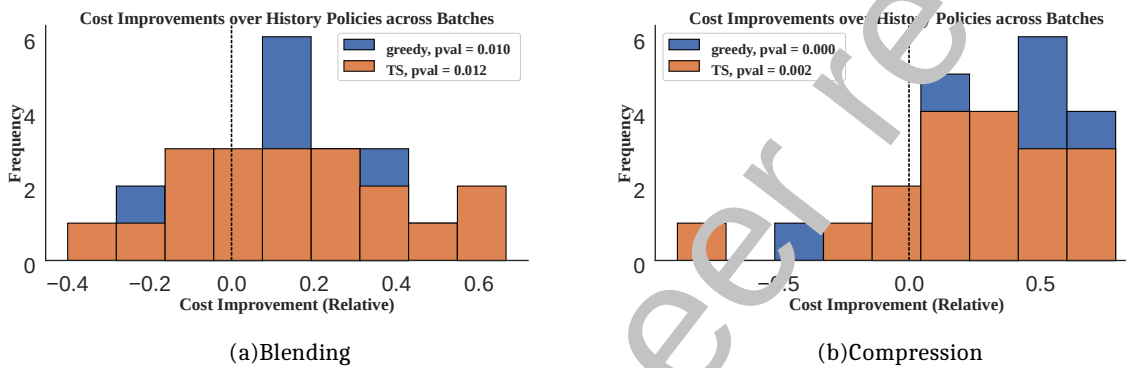


Figure 5 Improvements over the historical policy under the multi-task random forest representation

Finally, we examine Batch 42 (the second batch in the test period) and plot the cumulative costs within that batch as representative examples of learning performance under both known and unknown representations. We do not report the results for the compression stage here due to the limited number of time periods. In the blending stage, where the time horizon is long enough for the regret of the bandit algorithm to be relevant, we observe that both the greedy and TS algorithm significantly outperform historical policies while suffering small regret compared with the oracle policies. When B is unknown, our results highlight the advantage of learning a good representation: our proposed policies outperform the fixed representation counterparts, particularly as the time horizon t is large. This suggests that our proposed method effectively leverage online learning and a good adaptive representation to improve decision-making and reduce costs.

7. Conclusion

In this paper, we build a sequential contextual bandit model with related batches and propose a regret-minimizing algorithm by combining ideas from the multi-task learning

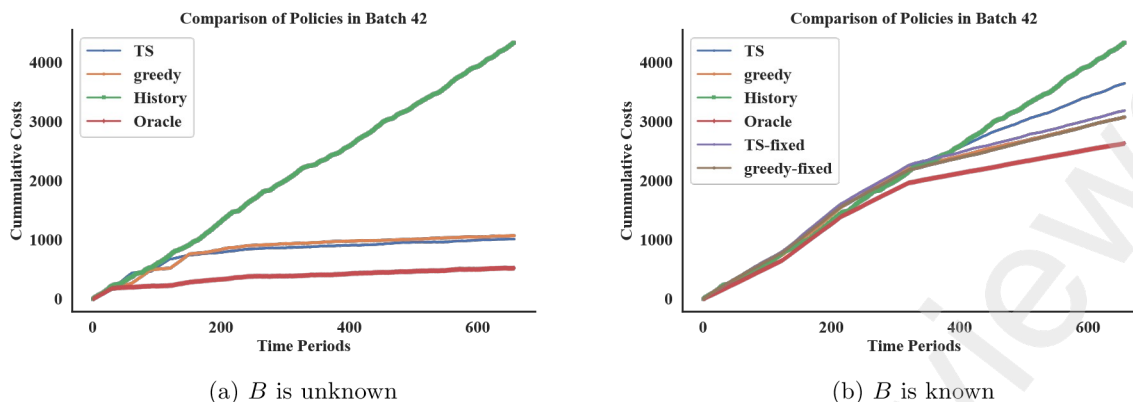


Figure 6 Cumulative costs in Batch 42 of the blending stage

and contextual bandit literature. This model is especially appropriate for the data in the MSOM challenge since we empirically show that batches of raw materials used in a continuous tablet manufacturing process are heterogeneous even when the observable properties of the raw materials are identical, i.e., the batches are related but not identical. We propose a multi-task random forest model to predict the cost of particular actions when the batches are related but heterogeneous. Our numerical studies validate that our proposed regret-minimization algorithm clearly outperforms the historical policies. Furthermore, the actions proposed by our proposed policy are supported by domain knowledge. Beyond the framework, the residence time distribution (RTD) of raw materials, after a time lag in the blending stage, can also be considered a state variable to predict the tablet quality in the compression stage. Incorporating this to enhance the prediction model is left for future work.

Beyond pharmaceutical tablet production, our framework can be applied to other domains when uncertainty evolves over time. We believe that the integration of multi-task learning with limited but relevant offline data presents exciting opportunities for advancing data-driven analytics in operations management.

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Online Appendix to Manufacturing Pharmaceutical Control

EC.1. Additional Results in Section 3

Interpretation of Individual Shapley value. For an individual data point $X_i = (X_i^{(1)}, \dots, X_i^{(d_x)})^\top$, the Shapley value of j -th feature $X_i^{(j)}$, $j \in [d_x]$ represents how much that feature contributes to shifting the prediction from the average prediction across all data points (Lundberg and Lee 2017). Mathematically: $f(X_i) - \mathbb{E}[f(X)]$ is the total deviation from the mean prediction and the deviation is distributed among features. The Shapley value for the j -th feature for X_i is:

$$\phi_j = \sum_{S: S \subset [d_x] \setminus \{j\}} \frac{|S|!(d_x - |S| - 1)!}{d_x!} [f^{S \cup \{j\}}(X_i) - f^S(X_i)], j \in [d_x],$$

where d_x is the number of features and S is a subset of features excluding the i -th one, f^S and $f^{S \cup \{i\}}$ are the model prediction using only features in S and $S \cup \{i\}$, which is a random forest with the subset of features.

In Figure EC.1, we report the Shapley value attribution results for other actions and find both environmental variables and raw material attributes lead to significant impacts on the action.

Heterogeneity. To demonstrate the heterogeneity of costs, we consider $K = 40$, i.e., batches in fitting the random forest, and visualize the 3D projection of $\{\theta_i^{(g)}\}_{i \in [K]}$ of them for blending and compression stages in Figure EC.2. We observe a wide range of differences in each stage.

EC.2. Proofs of Theorem 1 in Section 5

EC.2.1. Detailed Results in Section 5.1.1

Across the proofs, we define the sub-Gaussian norm of $X \in \mathbb{R}^d$ as:

$$\|X\|_{\psi_2} = \inf\{c > 0 : \mathbb{E}[\exp(\|X\|_2^2/c^2)] \leq 2\}.$$

The following matrix concentration inequality follows from standard results in Tropp (2011):

LEMMA EC.1 (Matrix Concentration Inequality). *For any $x_1, \dots, x_n$ are i.i.d. random vectors in $\mathbb{R}^p$ with $\|x_i\|_2 \leq R$ (or with sub-Gaussian norm R), and $\Sigma = \mathbb{E}[x_1 x_1^\top]$ and $\hat{\Sigma} = \frac{1}{n} \sum_{i=1}^n x_i x_i^\top$. Then there exists $C > 0$, such that for all $\delta > 0$, with probability at least $1 - \delta$:*

$$\|\hat{\Sigma} - \Sigma\|_{op} \leq CR^2 \sqrt{\frac{p + \log(1/\delta)}{n}}.$$

Based on Lemma EC.1, we provide the following concentration properties of the parameters of interest.

LEMMA EC.2 (Concentration Properties for Parameters). *Suppose Assumption 4 holds. Then $\forall k \in [K]$, with probability $1 - e^{-t}$ (where $t < 1$), we have: $\frac{p}{M+k-1} \sum_{j=1}^{M+k-1} \theta_j \theta_j^\top \succeq \frac{c_0}{2} \mathbf{I}$ for some constants c_0 . Furthermore, we have: $\sigma_{\min}(\Theta_{M+k-1}) \geq \alpha > 0$, where $\Theta_{M+k-1} = [\theta_1 | \dots | \theta_{M+k-1}] \in \mathbb{R}^{p \times (M+k-1)}$ and $\sigma_{\min}(\cdot)$ denotes the smallest positive singular value of a non-zero matrix.*

Proof of Lemma EC.2. Denote $S = \frac{1}{M+k-1} \sum_{j=1}^{M+k-1} \theta_j \theta_j^\top$. From Assumption 4, the sub-Gaussian norm of $\|\theta - \mu\|_2$ is bounded by $\sqrt{\lambda_{\max}(\Sigma)}$. Then from standard random matrix concentration inequalities in Lemma EC.1, with probability at least $1 - e^{-t}$, we have:

$$\|S - \mathbb{E}[S]\|_{op} \leq \lambda_{\max}(\Sigma) \sqrt{\frac{p+t}{M+k-1}} \leq C \lambda_{\min}(\Sigma) \sqrt{\frac{p+t}{M+k-1}}.$$

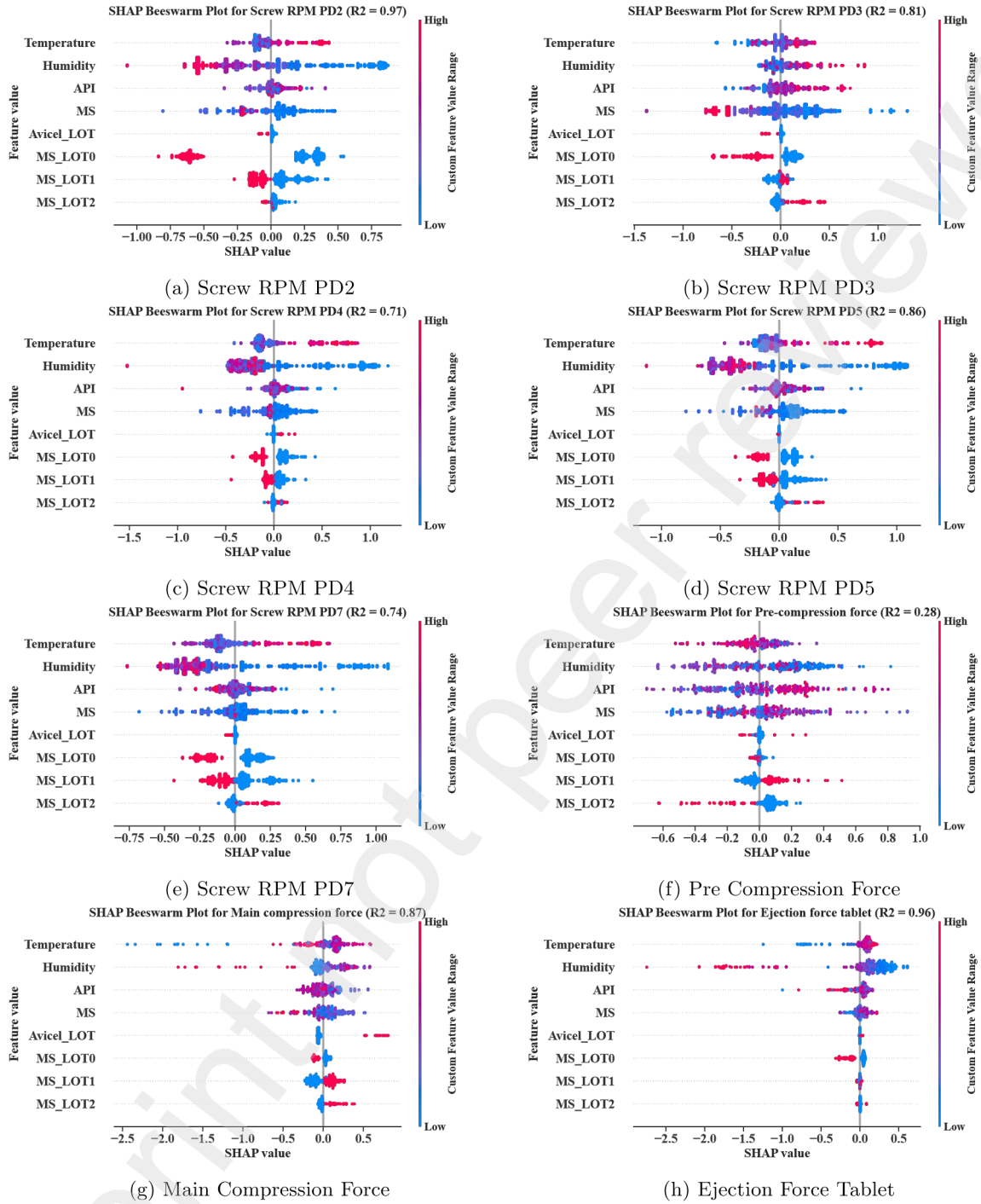
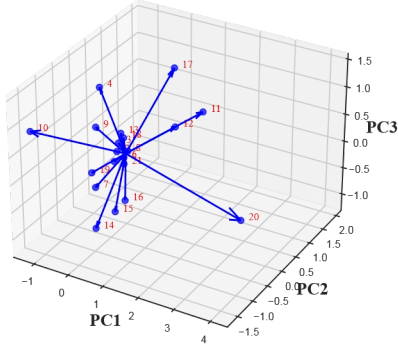


Figure EC.1 Shapley value distribution plots of actions given state variables.

Furthermore, note that $\mathbb{E}[S] = \Sigma + \mu\mu^\top$, the minimum eigenvalue satisfies:

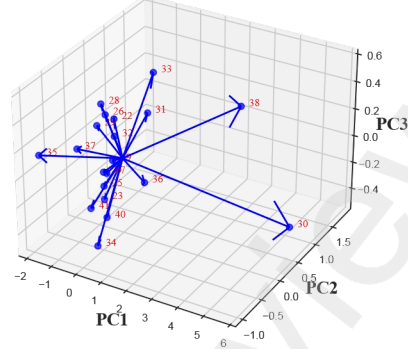
$$\lambda_{\min}(S) \geq \left(1 - \sqrt{\frac{p+t}{M+k-1}}\right) \lambda_{\min}(\Sigma) \geq \frac{\alpha}{2} \mathbf{I}.$$

Task-Specific Parameter Vectors of Feeder Cost Projected in 3D



(a) Blending

Task-Specific Parameter Vectors of Tablet_Press Cost Projected in 3D



(b) Compression

Figure EC.2 3D Projection visualization of heterogeneous effects of the random forest model to costs $\{\theta_i^{(g)}\}$

Multiplying the above inequality by p , we obtain:

$$\frac{p}{M+k-1} \sum_{j=1}^{M+k-1} \theta_j \theta_j^\top \succeq p \left(1 - \sqrt{\frac{p+t}{M+k-1}} \right) \alpha \mathbf{I} \succeq p \left(1 - \sqrt{\frac{p+t}{p+1}} \right).$$

For a Gaussian matrix $\Theta_{M+k-1} = [\theta_1 | \dots | \theta_{M+k-1}]$, recall Theorem 5.39 in Vershynin (2010) when we consider the matrix $\tilde{\Theta}_{M+k-1} = [\tilde{\theta}_1 | \dots | \tilde{\theta}_{M+k-1}] \in \mathbb{R}^{p \times (M+k-1)}$ with each $\tilde{\theta}_i \sim N(\mu, \mathbf{I})$. Then with probability $1 - \exp(-t)$, the smallest singular value of a centered Gaussian matrix is bounded by:

$$\sigma_{\min}(\tilde{\Theta}_{M+k-1}) \geq \sqrt{M+k-1} - C\sqrt{p} - t.$$

Comparing in our case, each column of Θ_{M+k-1} is drawn from $N(\mu, \Sigma)$ with $\lambda_{\min}(\Sigma) \geq \alpha$, then this results in: $\sigma_{\min}(\tilde{\Theta}_{M+k-1}) \geq (\sqrt{M+k-1} - C\sqrt{p} - t)\alpha \geq \alpha$, when $t < 1$ and $M \geq p+1$.

In the following, for each k , we define $G := [G_1 | \dots | G_{M+k-1}] \in \mathbb{R}^{d \times (M+k-1)}$ with G_j being the gradient of the true loss in the j -th task. That is:

$$G_j = \frac{1}{T} \sum_{t=1}^T \varepsilon_{j,t} \phi(x_{j,t}, s_{j,t}) \in \mathbb{R}^d. \quad (\text{EC.1})$$

We can represent $\mathcal{E}_{M+k-1}$ with G and G_j , which follows from a restate of Theorem A.5 from Duan and Wang (2023):

LEMMA EC.3 (Estimation Error). Suppose B is unknown and Assumptions 3 and 4 hold. Let $P^* \in \mathbb{R}^{d \times d}$ be the projection onto $\text{Range}(B)$. For each $k \in [K]$, let $(\hat{B}_{k-1}, \{\hat{\theta}_j\}_{j \in [M+k-1]})$ denote the solution of (9) in Algorithm 1 with the regularization parameter $\lambda > C \frac{M}{p\alpha^2} \max_{j \in [M+k-1]} \|G_j\|_2$ for some constant $C > 0$ in the optimization problem. Then the following bound holds, $\forall j \in [M+k-1]$:

$$\|\hat{B}_{k-1} \hat{\theta}_j - B \theta_j\|_2 \leq C \left(\frac{\max_{j \in [M+k-1]} \|P^* G_j\|_2}{\lambda_{\min}(H)} + \frac{\sqrt{p} \|G\|_2}{\alpha} \right),$$

where H is defined in Proposition 2.

LEMMA EC.4 (Sub-Gaussian Property of the Tail loss). Suppose Assumptions 2, 1, 3, 4 and 5 hold. Define $W = \sigma D / \sqrt{T}$. Then $\{G_i, i \in [M]\}$ is sub-Gaussian with the sub-Gaussian norm being W . Conditioned on $\mathcal{F}_k (k \geq 0)$, where $\mathcal{F}_k$ denotes the realization of $\{(s_{i,t}, x_{i,t}, c_{i,t})\}_{i,t=1}^{M+k,T}$ (including the historic data $\mathcal{D}$ and k new batches), then G_{M+k} is sub-Gaussian with the sub-Gaussian norm being W .

Proof of Lemma EC.4. For $i \in [M]$, the sub-Gaussian property of G_i directly follows from the fact that $\|\phi(x, s)\|_2 \leq D < \infty$ in Assumption 3. Conditioned on $\mathcal{F}_{0 \vee \{i-M\}}$, $\|\varepsilon_{j,t} \phi(x_{j,t}, s_{j,t})|_{\mathcal{F}_{0 \vee k}}\|_{\psi_2} \leq \sigma D$. Then the sub-Gaussian property holds for $i \geq M+k$. $\square$

LEMMA EC.5 (First-order Estimator Error). *Suppose Assumptions 2, 1, 3 and 4 hold. Following the notation in Lemma EC.3, for any $k \in [K]$, Then there are constants C_1 and C_2 such that the following hold with probability at least $1 - \exp(-t)$:*

- $\max_{j \in [M+k-1]} \|P^* G_j\|_2 \leq C_1 \sigma \sqrt{\frac{p + \log(M+k-1) + t}{T}};$
- $\|G\|_2 \leq \frac{C_2 \sigma}{\sqrt{T}} (\sqrt{d} + \sqrt{M+k-1} + \sqrt{t}).$

Note that when $k = 1$, the data fully relies on the offline data and becomes Theorem 5.39 in Vershynin (2010). Through Lemma EC.4, we generalize the result to the problem setup with both online and offline data, i.e., $k > 1$.

Proof of Lemma EC.5. Due to projection, we have: $\|P^* G_j\|_{\psi_2} \leq \|G_j\|_{\psi_2} \leq \sigma D \sqrt{\frac{1}{T}}$. By applying Theorem 2.1 in Hsu et al. (2012b), we have:

$$\mathbb{P} \left(\max_{j \in [M+k-1]} \|P^* G_j\|_2 > C_1 \sigma \sqrt{\frac{p + \log(M+k-1) + t}{T}} \right) \geq 1 - \exp(-t)/2.$$

Then we show the bound for $\|G\|_2$. We adapt the proof of Theorem 5.39 here. Here G is a $d \times (M+k-1)$ matrix with G_i being independent sub-Gaussian random variable in $\mathbb{R}^d$ with second moment $\Sigma = c\mathbf{I}$. This is because:

$$\begin{aligned} \mathbb{E}[G_i^\top G_j] &= \sum_{k=1}^d \mathbb{E}[G_{i,k} G_{j,k}] = \sum_{k=1}^d \mathbb{E}[G_{i,k}] \mathbb{E}[G_{j,k}] = 0 \text{ if } i, j \in [M], \\ \mathbb{E}[G_i^\top G_j] &= \mathbb{E}[G_i^\top G_j | \mathcal{F}_{j-M}] = 0, \text{ else } i < j, \end{aligned}$$

where the second equality follows from $\mathbb{E}[G_j | \mathcal{F}_{j-M}] = 0$ if $j > M$ and $G_i \in \mathcal{F}_{j-M}$ due to the martingale property.

Then setting $\delta = C \sqrt{\frac{d}{M+k-1}} + \frac{t}{\sqrt{M+k-1}}$, we need to show $\|\frac{1}{M+k-1} G^\top G - \mathbf{I}\| \leq \max\{\delta, \delta^2\}$, which is equivalent to showing:

$$\max_{u \in \mathcal{U}} \left| \frac{1}{M+k-1} \|Gu\|_2^2 - 1 \right| \leq \frac{\max\{\delta, \delta^2\}}{2}$$

for a $1/4$ -set $\mathcal{U}$ consisting the unit sphere S^{d-1} .

To conclude the proof, denote $V_i = G_i u$, and we only need to show: $\mathbb{P} \left(\left| \frac{1}{M+k-1} \sum_{i=1}^{M+k-1} V_i^2 - 1 \right| \geq \frac{\varepsilon}{2} \right) \leq 2 \exp(-\frac{c_1}{W^4} \min\{\varepsilon^2, \varepsilon\} T)$. Denote $S = \frac{1}{M+k-1} \sum_{i=1}^{M+k-1} V_i^2 - 1$, and each $V_i^2 - 1$ is sub-Exponential with parameters $4W^2$ (when $i > M$, the sub-Exponential property is with respect to $\mathcal{F}_{i-M}$). Then we have:

$$\begin{aligned} \mathbb{P}(S \geq t) &= \mathbb{P}(e^{\lambda S} \geq e^{\lambda t}) \leq e^{-\lambda t} \mathbb{E}[e^{\lambda S}] \\ &= e^{-\lambda t} \mathbb{E}[\exp(\frac{\lambda}{M+k-1} \sum_{i \in [M+k-1]} (V_i^2 - 1) | \mathcal{F}_{0 \vee (i-M)})] \\ &= e^{-\lambda t} \prod_i \mathbb{E}[\exp(\frac{\lambda}{M+k-1} \sum_{i \in [M+k-1]} (V_i^2 - 1))], \end{aligned}$$

where the first inequality follows by the tower property, and the fact that $V_i^2 - 1$ is a sub-Exponential random variable conditioned on $\mathcal{F}_{0 \vee (i-M)}$. The subsequent results following by the definition of the moment generating function of the sub-Exponential random variable. $\square$

Proof of Proposition 1. This result follows by plugging Lemma EC.5 into Lemma EC.3. $\square$

Proof of Proposition 2. Recall $F(\tilde{B}, \{\tilde{\theta}_j\}) = \frac{1}{T(M+k-1)} \sum_{t \in [T]} \sum_{j \in [M+k-1]} \mathbb{E}_{\varepsilon_{j,t}} \left[\left(c_{j,t} - \phi(x_{j,t}, s_{j,t})^\top \tilde{B} \tilde{\theta}_j \right)^2 \right]$.

We take the second-order derivative of $F(\tilde{B}; \{\tilde{\theta}_j\}_{j \in [M+k-1]})$ with respect to $\tilde{B}$ and $\{\tilde{\theta}_j\}_{j \in [M+k-1]}$ at $\tilde{B} = B$ and $\tilde{\theta}_j = \theta_j$. Then:

$$\begin{aligned} \frac{\partial^2 F}{\partial \tilde{B} \partial \tilde{B}} \Big|_{\tilde{B}=B} &= \frac{2}{T(M+k-1)} \sum_{t \in [T]} \sum_{j \in [M+k-1]} \theta_j \theta_j^\top \otimes \phi(x_{j,t}, s_{j,t}) \phi(x_{j,t}, s_{j,t})^\top, \\ \frac{\partial^2 F}{\partial \tilde{\theta}_j \partial \tilde{\theta}_j} \Big|_{\tilde{\theta}_j=\theta_j} &= \frac{2}{T(M+k-1)} \sum_{t \in [T]} B^\top \phi(x_{j,t}, s_{j,t}) \phi(x_{j,t}, s_{j,t})^\top B, \forall j \in [M+k-1]. \end{aligned}$$

The minimum eigenvalue of $\nabla^2 F_{\tilde{B}, \tilde{\theta}_j}$, is lower bounded by:

$$\lambda_{\min}(H) \geq \lambda_{\min} \geq \frac{2}{T(M+k-1)} \lambda_{\min}^\phi \min(\lambda_{\min}^B, \lambda_{\min}^\theta),$$

where:

$$\begin{aligned} \lambda_{\min}^\phi &= \min_{\|v\|_2=1} v^\top \left(\sum_{t,j} \phi(x_{j,t}, s_{j,t}) \phi(x_{j,t}, s_{j,t})^\top \right) v, \\ \lambda_{\min}^{B^*} &= \min_{\|w\|_2=1} w^\top (B^\top B) w, \\ \lambda_{\min}^{\theta^*} &= \min_{\|z\|_2=1} z^\top \left(\sum_{j \in [M+k-1]} \theta_j^* \theta_j^{*\top} \right) z. \end{aligned}$$

Following this result, taking a second-order Taylor expansion of $F(\hat{B}; \{\hat{\theta}_j\}_{j \in [M+k-1]}) - F(B; \{\theta_j\}_{j \in [M+k-1]})$ at the point of $B, \{\theta_j\}_{j \in [M+k-1]}$ (since F is strongly convex with respect to B and $\{\theta_j\}_{j \in [M+k-1]}$), we have:

$$F(\hat{B}; \{\hat{\theta}_j\}_{j \in [M+k-1]}) - F(B; \{\theta_j\}_{j \in [M+k-1]}) \geq \frac{\lambda_{\min}(H_{M+k-1})}{2} \left(\|B - \hat{B}\|_F^2 + \sum_{i \in [M+k-1]} \|\theta_i - \hat{\theta}_i\|_2^2 \right).$$

On the other hand, we have:

$$F(\hat{B}; \{\hat{\theta}_j\}_{j \in [M+k-1]}) - F(B; \{\theta_j\}_{j \in [M+k-1]}) \leq \lambda_{\max}(H_{M+k-1}) \max_{j \in [M+k-1]} \|\hat{B} \theta_j - B \theta_j\|_2^2 \leq \lambda_{\max}(H_{M+k-1}) \mathcal{E}_{M+k-1}^2.$$

Then we have:

$$\max\{\|B - \hat{B}\|_F^2, \|\theta_i - \hat{\theta}_i\|_2^2\} \leq \frac{\lambda_{\max}(H_{M+k-1})(\mathcal{E}_{M+k-1})^2}{\lambda_{\min}(H_{M+k-1})} = \kappa(H_{M+k-1})(\mathcal{E}_{M+k-1})^2. \quad \square$$

EC.2.2. Greedy

EC.2.2.1. Technical Lemmas In the context when no confusion arises, we use the single index for $\{(s_i, x_i, c_i)\}_{i \in [T]}$ to denote the state, action, and the cost; and the estimate for the $\theta, \hat{\theta}_i$ in the learning process. All of these subscripts of notations ignore the dependence with respect to the $(M+1)$ -th batch.

PROPOSITION EC.1 (Online Estimation Error of OLS under Misspecified Representation). For general $\hat{B} \neq B$ with Consider the data $\{(s_i, x_i, c_i)\}_{i \in [n]}$ where $\{(s_i, x_i)\}_{i \in [n]}$ are fixed, satisfying $c_i = \phi(x_i, s_i)^\top B\theta + \varepsilon_i$. Denote the following regression:

$$\begin{aligned}\hat{\theta}_n &= \arg \min_{\theta} \frac{1}{n} \sum_{i=1}^n (\phi(x_i, s_i)^\top \hat{B}\theta - c_i)^2, \\ \theta &= \arg \min_{\theta} \frac{1}{n} \sum_{i=1}^n \mathbb{E}_{\{\varepsilon_i\}_{i \in [n]}} (\phi(x_i, s_i)^\top B\theta - c_i)^2,\end{aligned}$$

where $\mathbb{E}_{\{\varepsilon_i\}_{i \in [n]}}[\cdot]$ is taken randomness over the noise. Denote $Z_{B,n} \in \mathbb{R}^{n \times p}$ (and $Z_{\hat{B},n} \in \mathbb{R}^{n \times p}$) as the matrix whose i -th row is $(B^\top \phi(x_i, s_i))$ (and $\hat{B}^\top \phi(x_i, s_i)$) respectively, $i \in [n]$. With probability $1 - \frac{1}{2} \exp(-t)$:

$$\|\hat{\theta}_n - \theta\|_2 \leq \sqrt{\frac{\sigma^2(p+2\sqrt{pt}+2t)}{\lambda_{\min}(Z_{B,n}^\top Z_{B,n})}} + \frac{tCD\|\hat{B} - B\|_F}{\lambda_{\min}(Z_{B,n}^\top Z_{B,n}/n)}. \quad (\text{EC.2})$$

Proof of Proposition EC.1. Denote $\hat{\theta}^*$ as the best-in class parameter under the misspecified representation $\hat{B}$, that is:

$$\hat{\theta}^* = \arg \min_{\theta} \frac{1}{n} \sum_{i=1}^n \mathbb{E}_{\{\varepsilon_i\}_{i \in [n]}} (\phi(x_i, s_i)^\top \hat{B}\theta - c_i)^2.$$

Denote $\Sigma = Z_{B,n}^\top Z_{B,n}/n$ and $\hat{\Sigma} = Z_{\hat{B},n}^\top Z_{\hat{B},n}/n$. Then we bound $\|\hat{\theta}_n - \theta\|_2 \leq \|\hat{\theta}_n - \hat{\theta}^*\|_2 + \|\hat{\theta}^* - \theta\|_2$.

The first term $\|\hat{\theta}_n - \hat{\theta}^*\|_2$ is related to the variance in the estimation of $\hat{\theta}_n$, from Remark 8 in Hsu et al. (2012a), we have: with probability at least $1 - \frac{1}{2} \exp(-t)$, $\|\theta_n - \hat{\theta}^*\|_\Sigma^2 \leq \frac{\sigma^2(p+2\sqrt{pt}+2t)}{n}$. Then we can bound $\|\hat{\theta}_n - \hat{\theta}^*\|_2 \leq \sqrt{\frac{\sigma^2(p+2\sqrt{pt}+2t)}{\lambda_{\min}(Z_{B,n}^\top Z_{B,n})}}$.

The second term $\|\hat{\theta}^* - \theta\|_2$ is related to the bias in the estimation $\hat{B}$. By plugging in the exact formula of θ and $\hat{\theta}^*$, with probability at least $1 - \frac{1}{2} \exp(-t)$, we have:

$$\begin{aligned}\|\hat{\theta}^* - \theta\|_2 &= \|(Z_{B,n}^\top Z_{B,n})^{-1} Z_{B,n}^\top \mathbf{c} - (Z_{\hat{B},n}^\top Z_{\hat{B},n})^{-1} Z_{\hat{B},n}^\top \mathbf{c}\|_\Sigma \\ &\leq \|(Z_{B,n}^\top Z_{B,n})^{-1}\| \|Z_{B,n}^\top - Z_{\hat{B},n}^\top\| \|\mathbf{c}\| + \|(Z_{B,n}^\top Z_{B,n})^{-1}\| \|Z_{B,n}^\top Z_{\hat{B},n} - Z_{\hat{B},n}^\top Z_{B,n}\| \|(Z_{\hat{B},n}^\top Z_{\hat{B},n})^{-1}\| \|\mathbf{c}\| \\ &\leq \frac{tC}{\lambda_{\min}(Z_{B,n}^\top Z_{B,n})/n} \|(\hat{B} - B) \sup_i \phi(x_i, s_i)\|_2 \leq \frac{tCD}{\lambda_{\min}(Z_{B,n}^\top Z_{B,n}/n)} \|\hat{B} - B\|_F.\end{aligned}$$

where $\mathbf{c} = (c_1, \dots, c_n)^\top$ and the first inequality follows by the identity $A^{-1} - B^{-1} = A^{-1}(B - A)B^{-1}$. $\square$

Then we present the following martingale inequality for matrix, which is extracted from Theorem 3.1 in Tropp (2011):

LEMMA EC.6 (Matrix Martingale Inequality). Consider $\mathcal{F}_1 \subset \mathcal{F}_2 \dots$ be a filtration and a finite sequence of positive semi-definite matrices $\{X_k\}$ with dimension d . Suppose $\lambda_{\max}(X_k) \leq R$ almost surely. Define the summation $Y := \sum_{k=1}^n X_k$ and $W := \sum_{k=1}^n \mathbb{E}[X_k | \mathcal{F}_{k-1}]$. Then for all $\mu \geq 0$, $\gamma \in [0, 1)$, we have:

$$\mathbb{P}(\lambda_{\min}(Y) \leq (1 - \gamma)\mu \text{ and } \lambda_{\min}(W) \geq \mu) \leq d \left(\frac{e^{-\gamma}}{(1 - \gamma)^{1-\gamma}} \right)^{\frac{\mu}{R}}.$$

Based on that, we can show the following matrix property hold:

LEMMA EC.7 (Minimum Eigenvalue Concentration). Recall $Z_{B,n}$ defined in Proposition EC.1 and suppose Assumption 5 holds. Then with probability $1 - \delta$, for any $T_0 \leq n \leq T$, $\lambda_{\min}(Z_{B,n}^\top Z_{B,n}) \geq \frac{n}{2}$, where $T_0 \geq \frac{4D^2 \log(pT/\delta)}{0.1c_1}$.

Proof of Lemma EC.7. We apply matrix concentration inequality in Lemma EC.6 to the sequence $\{B^\top \phi(x_k, s_k)(B^\top \phi(x_k, s_k))^\top\}_{k \in [T]}$. From Assumption 5, we have: $\lambda_{\min}(\mathbb{E}[B^\top \phi(x_{k+1}, s_{k+1})(B^\top \phi(x_{k+1}, s_{k+1}))^\top | \mathcal{F}_k]) \geq \min_x \lambda_{\min}(\mathbb{E}[B^\top \phi(x, s)(B^\top \phi(x, s))^\top]) \geq c_1$. Thus, by setting $Y = \sum_{k \in [T]} B^\top \phi(x_k, s_k)(B^\top \phi(x_k, s_k))^\top = Z_{B,n}^\top Z_{B,n}$, $W = \mathbb{E}[B^\top \phi(x_{k+1}, s_{k+1})(B^\top \phi(x_{k+1}, s_{k+1}))^\top | \mathcal{F}_k]$, $\gamma = 1/2$, $R = D$, $\mu = \frac{c_1 n}{2}$ in Lemma EC.6, we have:

$$\mathbb{P}\left\{\lambda_{\min}(Z_{B,n}^\top Z_{B,n}) \leq \frac{c_1 n}{2}\right\} \leq p\left(\sqrt{2}e^{-0.5}\right)^{\frac{c_1 n}{4D^2}} = \exp(\log p - \frac{0.1c_1 n}{4D^2}).$$

Applying the above with a union bound for every $T_0 \leq n \leq T$, with probability at least $1 - \delta$, we have: $\lambda_{\min}(Z_{B,n}^\top Z_{B,n}) \geq \frac{c_1 n}{2}$ for any $n \in [T_0, T]$. $\square$

EC.2.2.2. $K = 1$. In the proof of the greedy algorithm, we directly consider the case when B is unknown while leveraging (EC.2) in Proposition EC.1. Note that when B is known, the proof follows the same while setting $B = \hat{B}$ in (EC.2). In the following, denote $\hat{x}_i$ as the action taken by Algorithm 2 and x_i^* at each time $i \in [T]$.

$$\begin{aligned} \mathbb{E}[R_{1,T}] &= \sum_{i=1}^T (\phi(\hat{x}_i, s_i)^\top B\theta - \phi(x_i^*, s_i)^\top B\theta) \\ &= \sum_{i=1}^T (\phi(\hat{x}_i, s_i)^\top B\theta - \phi(\hat{x}_i, s_i)^\top \hat{B}\theta) + \sum_{i=1}^T (\phi(x_i^*, s_i)^\top \hat{B}\theta - \phi(x_i^*, s_i)^\top B\theta) \end{aligned} \quad (\text{EC.3})$$

$$+ \sum_{i=1}^T (\phi(\hat{x}_i, s_i)^\top \hat{B}\theta - \phi(x_i^*, s_i)^\top \hat{B}\theta), \quad (\text{EC.4})$$

where the error in (EC.3) arises due to the representation error and becomes zero when $B = \hat{B}$. The error in (EC.4) arises due to the estimation error from $\hat{\theta}$.

We bound them one by one. First, consider the following event:

$$\mathcal{A} = \left\{ \begin{aligned} &\|\hat{\theta}_n - \theta\|_2 \leq \sqrt{\frac{\sigma^2(p+2\sqrt{p \log T} + 2 \log T)}{c_1 n}} + \frac{CD \log T \|\hat{B} - B\|_F}{c_1}, \quad \forall T_0 \leq n \leq T, \\ &\lambda_{\min}(Z_{B,n}^\top Z_{B,n}) \geq \frac{c_1 n}{2}, \quad \forall T_0 \leq n \leq T, \\ &\mathcal{E}_M \leq \sqrt{\frac{d \log T}{M \alpha T}}, \\ &\kappa(H_M) \leq \kappa < \infty \end{aligned} \right\}.$$

Within $\mathcal{A}$, based on Proposition EC.1 for the first part, Lemma EC.7 for the second and fourth part, Proposition 1 for the third part, we know: $\mathbb{P}(\mathcal{A}_0) \geq 1 - C/T$ for some constant C .

Under the event of $\mathcal{A}$, for the term in (EC.3), following Proposition 2, we have:

$$\begin{aligned} &\sum_{i=1}^T (\phi(\hat{x}_i, s_i)^\top B\theta - \phi(\hat{x}_i, s_i)^\top \hat{B}\theta) + \sum_{i=1}^T (\phi(x_i^*, s_i)^\top \hat{B}\theta - \phi(x_i^*, s_i)^\top B\theta) \\ &\leq 2\|B - \hat{B}\|_F \|\theta\|_2 \max_{x,s} \|\phi(x, s)\|_2 \\ &\leq 2D\|\theta\|_2 T \mathcal{E}_M \sqrt{C} \leq 2D\|\theta\|_2 p \sqrt{\frac{dT}{M \alpha}}. \end{aligned}$$

Therefore, $\mathbb{E}\left[\sum_{i=1}^T (\phi(\hat{x}_i, s_i)^\top B\theta - \phi(\hat{x}_i, s_i)^\top \hat{B}\theta) + \sum_{i=1}^T (\phi(x_i^*, s_i)^\top \hat{B}\theta - \phi(x_i^*, s_i)^\top B\theta)\right] \leq 2D\|\theta\|_2 p \sqrt{\frac{dT}{M \alpha}} + D.$

For the term in (EC.4), for each $i \in [T]$, we have:

$$\begin{aligned} \phi(\hat{x}_i, s_i)^\top \hat{B}\theta - \phi(x_i^*, s_i)^\top \hat{B}\theta &\leq \left(\phi(\hat{x}_i, s_i)^\top \hat{B}\theta - \hat{\phi}(\hat{x}_i, s_i)^\top \hat{B}\theta \right) + \left(\hat{\theta}_i^\top \hat{B}\phi(x_i^*, s_i) - \theta_i^\top \hat{B}\phi(x_i^*, s_i) \right), \\ &\leq T \|\hat{\theta}_i - \theta\|_2 \|\hat{B}\|_F (\|\phi(\hat{x}_i, s_i)\|_2 + \|\phi(x_i^*, s_i)\|_2) \\ &\leq 2D \|\hat{\theta}_i - \theta\|_2 (\|B\|_F + \sqrt{\kappa} \mathcal{E}_M) \\ &\leq 4D^2 \|\hat{\theta}_i - \theta\|_2, \end{aligned}$$

where the first inequality follows from the optimality of the Greedy algorithm, that is: $\phi(\hat{x}_i, s_i)^\top \hat{B}\hat{\theta}_i \leq \phi(x_i^*, s_i)^\top \hat{B}\hat{\theta}_i, \forall i \in [T]$ and the last inequality holds since $\sqrt{\kappa(H_M)}\mathcal{E}_M \leq \|B\|_F$ when $T \geq T_0$.

Summing them from 1 to T , we have:

$$\begin{aligned} &\mathbb{E} \left[\sum_{i=1}^T \left(\phi(\hat{x}_i, s_i)^\top \hat{B}\theta - \phi(x_i^*, s_i)^\top \hat{B}\theta \right) \right] \\ &\leq \mathbb{E} \left[4D^2 \sum_{i=1}^T \|\hat{\theta}_i - \theta\|_2 \right] \\ &\leq \mathbb{E} \left[4D^2 \sum_{n=T_0}^T \left(\sqrt{\frac{\sigma^2(p + 2\sqrt{p} \log T + 2 \log T)}{c_1 n}} + \frac{CD \log T \|\hat{B} - B\|_F}{c_1} \right) |\mathcal{A}| \right] \\ &\quad + (\mathbb{P}(\mathcal{A}) 2DT + T_0) \left(\|B\|_F + \frac{L\mathcal{E}_m}{\zeta} \right) \\ &\leq \mathbb{E} \left[4D \|B\|_F \sum_{n=T_0}^T \left(\sqrt{\frac{\sigma^2(p + 2\sqrt{p} \log T + 2 \log T)}{c_1 n}} + \frac{CD \log T \|\hat{B} - B\|_F}{c_1} \right) |\mathcal{A}| \right] + \Theta(1) \\ &\leq 4D^2 \left(\frac{\sigma \sqrt{p} (\log T)^{1/2}}{\sqrt{c_1}} \sum_{n=T_0}^T \sqrt{\frac{1}{n}} + T \frac{\kappa CD \mathcal{E}_M}{c_1} \right) \\ &\leq 4D^2 \frac{\sqrt{pT \log T}}{\sqrt{c_1}} + \frac{4pD^2 C}{\alpha c_1} \left(\sqrt{\frac{dT}{M}} + \sqrt{T} \right) = \tilde{\Theta} \left(\left(1 + \sqrt{\frac{d}{M\alpha}} \right) p\sqrt{T} \right), \end{aligned}$$

The third inequality follows from Assumption 5 and Lemma EC.7 such that $\mathbb{P}(\mathcal{A}) \geq 1 - C/T$. The last equality ignores the constant with respect to D and α .

Therefore, by combining the result in (EC.3) and (EC.4), we have:

$$\mathbb{E}[R_{1,T}] \leq \tilde{O} \left(\left(1 + \sqrt{\frac{d}{M\alpha}} \right) p\sqrt{T} \right).$$

As a special case when B is known, $\hat{B} = B$, we repeat the previous arguments while eliminating the representation error (e.g., Equation (EC.3) becomes 0). This leads to the error bound of $\tilde{O}(p\sqrt{T})$. $\square$

EC.2.2.3. $K > 1$. When B is unknown, following the same analysis as in Appendix EC.2.2.2, for the k -th “test” batch, the expected regret is upper bounded by $(1 + \sqrt{\frac{d}{\alpha(M+k-1)}})p\sqrt{T}$.

Denote the regret at each new online round k as R_k . Following, we sum the total regret across K batches. This gives:

$$\mathbb{E}[R_{K,T}] = \sum_{k=1}^K \left(1 + \sqrt{\frac{d}{\alpha(M+k-1)}} \right) p\sqrt{T}$$

$$= Kp\sqrt{T} + p\sqrt{\frac{d}{\alpha}} \left(\sqrt{1 + \frac{K}{M}} - 1 \right) \sqrt{T} \leq Kp\sqrt{T} + p\sqrt{\frac{dKT}{M\alpha}},$$

where the second equality follows from $\sum_{k=1}^K \sqrt{\frac{d}{M+k-1}} \leq \sqrt{\frac{d}{M}} \int_0^K \frac{dt}{\sqrt{1+\frac{t}{M}}} \leq 2\sqrt{d} \left(\sqrt{1 + \frac{K}{M}} - 1 \right)$, and the first inequality follows $\sqrt{1+a} \leq 1 + \sqrt{a}$ for any $a > 0$.

When B is known, the sum of regret $\mathbb{E}[R_{K,T}] = K\mathbb{E}[R_{1,T}] \leq pK\sqrt{T}$. $\square$

EC.2.3. Thompson Sampling

EC.2.3.1. Technical Lemmas We directly consider the case when B is unknown. When B is known, the result directly follows by replacing $\mathcal{E}_{M+k-1}$ with the standard error in regression for $\|\hat{\theta}_j - \theta_j\|_2$.

DEFINITION EC.1 (GENERALIZED n -MONTE CARLO). We say a family of Bayesian contextual algorithms $\{\text{alg}(w) : w \in \mathcal{W}\}$ satisfy the generalized n -Monte Carlo property if, for all possible trajectories τ_h ,

$$\text{TV}(P_{\text{alg}(\theta)}(a_h|\tau_{h-1}) \parallel P_{\text{alg}(\theta')}(a_h|\tau_{h-1})) \leq \text{TV}(P_\theta(\phi|\tau_{h-1}) \parallel P_{\theta'}(\phi|\tau_{h-1})).$$

DEFINITION EC.2 (TAIL EXPECTATION). Let X be a nonnegative random variable on a probability space $(\Omega, \mathcal{F})$ with law P and finite expectation $\mathbb{E}[X] < \infty$. We define its tail expectation, as a function of probabilities $p \in (0, 1]$ as:

$$\Psi_X(p) := \frac{1}{p} \sup_Y \{\mathbb{E}[XY], \text{ s.t. } Y : (\Omega, \mathcal{F}) \mapsto [0, 1], \text{ and } \mathbb{E}[Y] \leq p\}.$$

For $p > 1$, we extend $\Psi_X(p) = \mathbb{E}[X]$. Overloading notation, we let $\Psi_w(p)$ denote the upper tail function over $\text{diam}(\mu)$ when drawn from $\mathbb{P}_w$:

$$\Psi_w(p) := \Psi_{\text{diam}(\mu)}(p) \text{ where } \mu \sim \mathbb{P}_w.$$

We extract Theorem E.1 in Simchowitz et al. (2021) (with a bandit special example) as follow, where we denote $\text{alg}(w)$ as the algorithm with prior w :

LEMMA EC.8 (Cost Difference under Different Prior). *Consider $\text{alg}(w)$ satisfy the n -Monte Carlo property on the time period T with $\{(s_t, x_t(\tilde{w}), c_t(\tilde{w}))\}_{t \in [T]}$ as the state, action and costs generated by $\text{alg}(\tilde{w})$ across the time horizon T . Denote $R_w(\text{alg}(w'))$ as the total expected cost for the algorithm with prior w' on the whole time horizon T where the true prior of the unknown parameter is w' . Then for $\varepsilon = \text{TV}(\mathbb{P}_w, \mathbb{P}_{w'})$:*

$$|R_w(\text{alg}(w)) - R_w(\text{alg}(w'))| \leq 4nT^2\varepsilon \cdot \bar{\Psi}_{w'}(2T\varepsilon),$$

where $\bar{\Psi}_w(p) = \Psi_{|\bar{c}_{T,w}|}(p)$ with $\bar{c}_{T,w} := \frac{1}{T} \sum_{t=1}^T c_t(\text{alg}(w))$.

Proof of Lemma 2. We know Algorithm 3 is a 1-Monte Carlo algorithm from Simchowitz et al. (2021). Then plugging the result of Lemma EC.8 in into our problem instance, we have:

$$R_{1,T}(w) - R_{1,T}(w') \leq 4T^2\varepsilon \bar{\Psi}_{w'}(2T\varepsilon) \leq 4T^2\varepsilon (\|\phi(x, s)\|_\infty B\theta + \frac{\sigma}{\sqrt{T}}) \leq CT^2\varepsilon,$$

where the second inequality follows from Lemma B.6 in Simchowitz et al. (2021) and:

$$|\bar{c}_{T,w}| = \left| \frac{1}{T} \sum_{t=1}^T (\varepsilon_t + \phi(x_t, s_t)^\top B\theta) \right| \leq \|\phi(x, s)\|_\infty B\theta + \left| \frac{1}{T} \sum_{t=1}^T \varepsilon_t \right|.$$

Since $\frac{1}{T} \sum_{t=1}^T \varepsilon_t$ is sub-Gaussian with the parameter $\frac{\sigma}{\sqrt{T}}$, that with probability $1 - e^{-t}$ conditioned on the randomness of $\{\varepsilon_t\}_{t \in [T]}$, $\bar{\Psi}_{w'}(2T\varepsilon) \leq \|\phi(x, s)\|_\infty B\theta + \frac{\sigma\sqrt{t}}{\sqrt{T}}$. Integrating over t yields the above result. $\square$

LEMMA EC.9 (TV Distance Bound). *For two Gaussian distributions $N(\mu_1, \Sigma_1)$ and $N(\mu_2, \Sigma_2)$ supported in $\mathbb{R}^d$, then:*

$$\begin{aligned} TV(N(\mu_1, \Sigma_1), N(\mu_2, \Sigma_2)) &\leq \frac{1}{2} \sqrt{\text{tr}(\Sigma_1^{-1} \Sigma_2 - I) + (\mu_1 - \mu_2)^\top \Sigma_1^{-1} (\mu_1 - \mu_2) - \log \det(\Sigma_2 \Sigma_1^{-1})}. \\ &\leq C(\|\mu_1 - \mu_2\|_2 + \|\Sigma_1 - \Sigma_2\|_{op}) \end{aligned} \quad (\text{EC.5})$$

EC.2.3.2. $K = 1$. To prove the regret of Algorithm 1 using the component of Algorithm 3, from Lemma 2, we only need to bound the prior difference for each k . Firstly, we consider $k = 1$ and $\hat{\mu}, \hat{\Sigma}$ then denote the empirical mean and covariance of parameters in the M batches and $\hat{B} = \hat{B}_0$.

Consider $\mathbb{P}_w$ as the true Gaussian distribution $N(B\mu, B\Sigma B^\top)$ with parameters B, μ, Σ ; $\mathbb{P}_{\hat{w}}$ as the estimated Gaussian distribution $N(\hat{B}\hat{\mu}, \hat{B}\hat{\Sigma}\hat{B}^\top)$ estimated prior with parameters $\hat{B}, \hat{\mu}, \hat{\Sigma}$. We bound the misspecification error between the true prior $\mathbb{P}_w = N(B\mu, B\Sigma B^\top)$ and the estimated prior $\mathbb{P}_{\hat{w}} = N(\hat{B}\hat{\mu}, \hat{B}\hat{\Sigma}\hat{B}^\top)$. From Lemma EC.9, we only need to control the mean and covariance difference.

For the mean difference, with probability $1 - e^{-t}$ with respect to all the randomization, we have:

$$\begin{aligned} \|\hat{B}\hat{\mu} - B\mu\|_2 &= \left\| \frac{\hat{B}}{M} \sum_{i \in [M]} \hat{\theta}_i - \frac{B}{M} \sum_{i \in [M]} \theta_i + \frac{B}{M} \sum_{i \in [M]} \theta_i - B\mu \right\|_2 \\ &\leq \left\| \frac{\hat{B}}{M} \sum_{i \in [M]} \hat{\theta}_i - \frac{B}{M} \sum_{i \in [M]} \theta_i \right\|_2 + \left\| \frac{B}{M} \sum_{i \in [M]} \theta_i - B\mu \right\|_2 \\ &\leq \|\hat{B} - B\|_F \left\| \frac{1}{M} \sum_{i \in [M]} \hat{\theta}_i \right\|_2 + \|B\|_F \left(\max_{i \in [M]} \|\hat{\theta}_i - \theta_i\|_2 + \left\| \mu - \frac{1}{M} \sum_{i \in [M]} \theta_i \right\|_2 \right) \\ &\leq \|B\|_F \left(p \sqrt{\frac{d}{M\alpha T}} + \frac{p}{\sqrt{M}} \right) \lesssim \|B\|_F p \sqrt{\frac{d}{M\alpha T}}, \end{aligned}$$

where the third inequality follows from the fact both $\|\hat{B} - B\|_F, \max_{i \in [M+k-1]} \{\|\hat{\theta}_i - \theta_i\|_2\} \leq Cp\sqrt{\frac{dt}{M\alpha T}}$ from Proposition 2 and $\left\| \frac{1}{M} \sum_{i \in [M]} (\theta_i - \mu) \right\|_2 \leq p\sqrt{\frac{t}{M}}$. And the last inequality follows from Assumption 4.

For the covariance difference, similarly, we can bound:

$$\|\hat{B}\hat{\Sigma}\hat{B} - B\Sigma B^\top\|_{op} \leq \|B\|_F \|\Sigma\|_{op} (\|\hat{\Sigma} - \Sigma\|_{op} + \|\hat{B} - B\|_F),$$

where $\|\hat{\Sigma} - \Sigma\|$ is further bounded as follows:

$$\|\hat{\Sigma} - \Sigma\|_{op} \leq \frac{1}{M} \sum_{i=1}^M \|\hat{\theta}_i - \theta_i\|_2 + \frac{1}{M} \sum_{i=1}^M \|\theta_i - \mu\|_2 \leq p\sqrt{\frac{d}{M\alpha T}} + p\sqrt{\frac{1}{M}}.$$

Therefore, we also have: $\|\hat{B}\hat{\Sigma}\hat{B} - B\Sigma B^\top\|_{op} \leq p\sqrt{\frac{d}{M\alpha T}}$.

Combining them together, we have:

$$\begin{aligned} R_{1,T}(w') &= R_{1,T}(w) + R_{1,T}(w') - R_{1,T}(w) \\ &\leq p\sqrt{T} + T^2 \cdot TV(N(\hat{B}\hat{\mu}, \hat{B}\hat{\Sigma}\hat{B}^\top), N(B\mu, B\Sigma B^\top)) \leq p\sqrt{T} + p\sqrt{\frac{dT^3}{M\alpha}}. \end{aligned}$$

where the second inequality follows from the regret of Thompson sampling under the true prior is $O(p\sqrt{T})$ from Russo and Van Roy (2014).

EC.2.3.3. $K > 1$. Following the The expected k -th round regret is bounded by $\sqrt{\frac{dT^3}{(M+k-1)\alpha}}$. Then summing them up among all “test” batches from 1 to K , we have: $E[R_{K,T}] = \sum_{k=1}^K \sqrt{\frac{dT^3}{(M+k-1)\alpha}} = pK\sqrt{\frac{dT^3}{M\alpha}}$.

EC.3. Additional Experimental Details in Section 6

EC.3.1. Data Selection and Preprocessing

We discussed the data extraction from procedure different sources in the dataset, with details referred to Martagan et al. (2024). For the state variables, we described data from the following two sources:

Temperature and Humidity: We include the temperature and humidity from the records, which are measured every 15 minutes.

Raw Material: We consider some raw material attributes, especially those related to raw material that vary across batches of new material. Specifically, we focus on the amount of API, the amount of Magnesium Stearate, and the lot (bag) of Magnesium Stearate. The amount of API and Magnesium are included since they change with each refill, and the lot (bag) of Magnesium Stearate is also considered due to its limited yet meaningful variability (4 different lots across 57 refills). Although the refill frequencies between Magnesium Stearate and API differs slightly, we align them according to the refill time of API. As mentioned in the main body, the time horizon between consecutive API refill typically last 1-2 hours.

Note that there are two different lots of Avicel PH102. However, one lot is used predominantly with Avicel PH102 being fed in six consecutive hours, while most of the raw materials are fed from the other lot at different times, resulting in significant data imbalance. Besides, we do not observe significant differences between these two lots in terms of the action and cost performance from Section 3.

As described in Figure 2, these state variables are used in both blending and compression stages.

For the **blending** stage, we extract the actions and the cost from the following data sources:

LiW-Feeder Data: We extract the LiW feeders screws speed as the action to adjust, which controls the corresponding mass flow of each feeder. Each entry is measured every second.

Blender Data: We set the cost of the screws speed as the summation of total loss in Table 1, i.e., for each time:

$$\text{Cost} = \sum_{i \in \{1, 2\}} \text{MassflowBlender}_i \times (\text{PotencyBlender}_i - 100 + \text{OOSConcentration}_i). \quad (\text{EC. 6})$$

This controls the total quantity loss multiplied by their quality. Each entry is measured every second.

Therefore, for each batch i , the total time length of the dataset $\{(S_{i,t}, x_{i,t}, c_{i,t})\}_{t \in [T]}$ is 1-2 hours, with each t partitioned by second. We filter the dataset by dropping the values where some feeder(s) do not work, i.e., when some coordinates of $x_{i,t}$ is non-positive.

For the **compression** stage, we extract the actions and the cost from the following data sources:

Tablet Press: We choose the precompression force, main compression force, ejection force of tablet, and compression cycle fill depth as four actions since they usually control the quality attribute of the tablet. Each entry is measured every second.

Tablet Properties: We set the cost of the compression steps as the absolute deviation of average hardness against the optimal quality in the golden profile, 14.5 (Holman et al. 2021). We do not choose other variables since other quality attributes, i.e., weight and thickness, are of less variability. Each entry is averaged over ten results, around 15 minutes.

Finally, we normalize the variables in each stage by scaling each variable of (s, x, c) in each dataset by its mean and standard deviation.

EC.3.2. Experimental Details in Section 6.2

We measure the performance of the poly and forest model across each training criterion via tuning the following hyperparameters:

- MTL-Forest and Pool-Forest: tree number d in $\{5, 10, 20\}$, max depth of each tree in $\{2, 3, 5\}$;
- Ind-Forest: tree number d in $\{2, 3, 5\}$, max depth of each tree in $\{3, 5\}$;
- MTL-Poly: the representation dimension $p \in \{5, 10, 20\}$, poly feature degree in $\{1, 2, 3\}$;
- Pool-Poly and Ind-Poly: poly feature degree in $\{1, 2, 3\}$ since there are no representation B involved.

In each forest model, we fix the max feature as 0.5, that is, we use 50% of all the features randomly for each tree.

EC.3.3. Experimental Details in Section 6.3

We detail the optimization steps here. Note that across Algorithms 2 and 3, at each time period, we are facing the following optimization problem: Given each state s , we obtain the optimal action by:

$$a^* \in \arg \min_{x \in \mathcal{X}} \phi(s, x)^\top B \theta. \quad (\text{EC.7})$$

In $\mathcal{X}$, we set each component of the decision as follows:

$$\mathcal{X} = \{x \in \mathbb{R}^{d_x} \mid L_i \leq x_i \leq U_i, i \in [d_x]\}. \quad (\text{EC.8})$$

For $i \in [d_x]$, we choose 40% and 60% quantile across the decision range for L_i and U_i to eliminate the effects of extreme values. which is a reasonable decision boundary that does not deviate from historic policy and has the potential to outperform existing benchmarks. This also reflects the reality that we do not want extreme actions that may lead to changes in temperature or humidity.

For each optimization problem in (EC.7), given the complexity of the objective function – whether modeled by a random forest (or a third-order polynomial function), we estimate the solution using a random search approach. Specifically, we randomly select 1000 points within the feasible region defined by (EC.8) and evaluate the best one. The performance and their observed trends do not change when varying the number of search points, demonstrating the robustness of this random search procedure.

EC.3.4. Additional Results in Section 6.3

We report the improvement over the history policy for greedy and TS algorithms with different random splits for the random forest model in Figure EC.3. In most cases, greedy and TS algorithms achieve significantly better performance than the history policy, around 20% improvements.

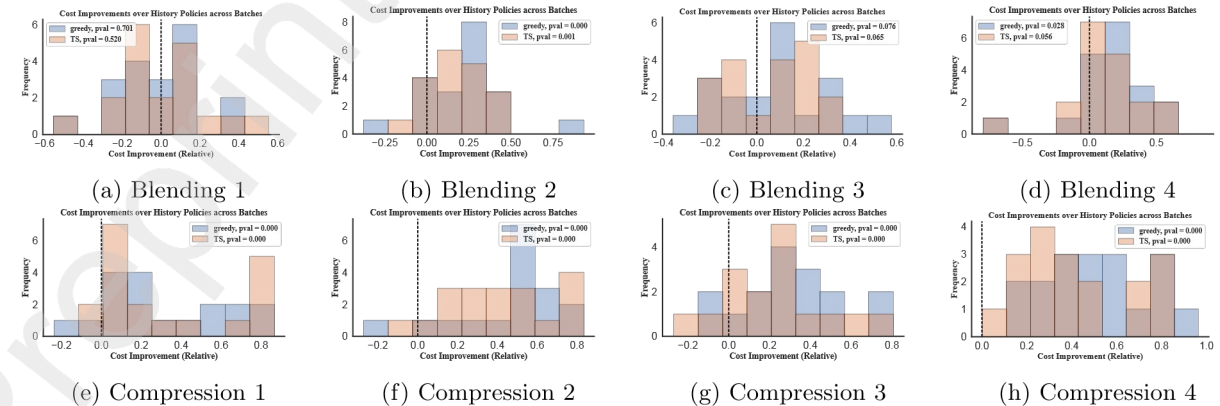


Figure EC.3 Improvements over the history policy under the multi-task random forest over other random splits.